

Cell Balancing

7.1 Introduction

Cell balancing in BQ78350-R1A is accomplished by connecting an external parallel bypass load to each cell of the associated AFE, and enabling the bypass load depending on each cell's charge state. The bypass load is typically formed by a P-CH MOSFET and a resistor connected in series across each battery cell. The filter resistors that connect the cell tabs to VC1~VC15 pins of the associated AFE are required to be 1 k Ω .

Using this circuit, the BQ78350-R1A balances the cells during charge by enabling the bypass around those cells above the threshold set in **Cell Balance Threshold** if the maximum difference in cell voltages exceeds the value programmed in **Cell Balance Min**. During cell balancing, the BQ78350-R1A measures the cell voltages at an interval set in **Cell Balance Interval**.

The cell(s) to be balanced are prioritized by highest cell voltage but the BQ78350-R1A will not try to balance adjacent cells. If adjacent cells need to be balanced, the BQ78350-R1A will alternate between the highest and next-highest adjacent cells until they are balanced.

On the basis of the cell voltages, the BQ78350-R1A either selects the appropriate cell to discharge or adjusts the cell balance threshold up by the value programmed in **Cell Balance Window** when all cells exceed the cell balance threshold or the highest cell exceeds the cell balance threshold by the cell balance window.

More in-depth details and data on this cell balancing algorithm can be found in:
<http://www.ti.com/lit/slua155>.

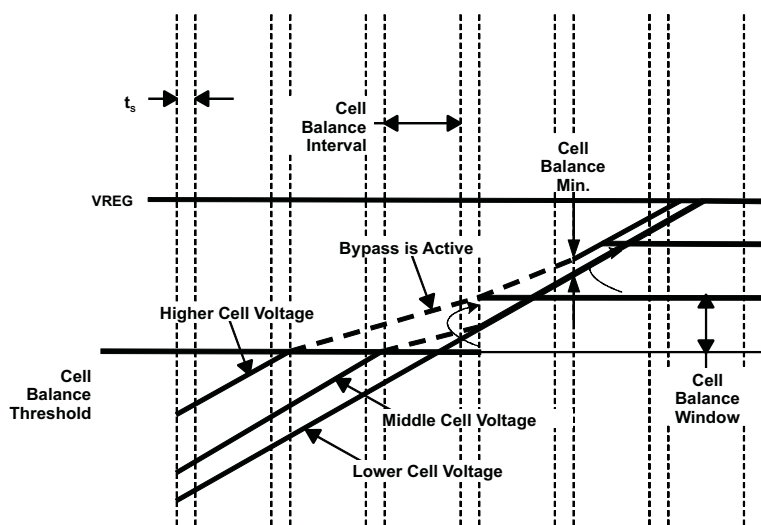


Figure 7-1. Cell Balancing

Cell balancing only occurs when charging current is detected, and on non-adjacent cells at the same time. The cell balance threshold is reset to the value in Cell Balance Threshold at the start of every charge cycle. The threshold is only adjusted once during any balance interval.

The configuration data flash is stored in **Charge Algorithm: Cell Balancing Config**.

7.1.1 Cell Balancing Configuration

Class	Subclass	Name	Type	Min	Max	Default	Unit
Charge Algorithm	Cell Balancing Config	Cell Balance Threshold	I2	0	5000	3900	mV
Charge Algorithm	Cell Balancing Config	Cell Balance Window	I2	0	5000	100	mV
Charge Algorithm	Cell Balancing Config	Cell Balance Min	U1	0	255	40	mV
Charge Algorithm	Cell Balancing Config	Cell Balance Interval	U1	0	255	20	s

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	Balancing Configuration	Hex	1	0x00	0xFF	0x01	—

7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	CB

RSVD (Bits 7–1): Reserved

CB (Bit 0): Cell balancing

1 = Enabled (default)

0 = Disabled

Power Modes

8.1 Introduction

To enhance battery life, the BQ78350-R1A supports several power modes to minimize power consumption during operation.

8.2 NORMAL Mode

In NORMAL mode, the device takes voltage, current, and temperature readings every 250 ms, performs protection and gauging calculations, updates SBS data, and makes status readings at 1-s intervals. Between these periods of activity, the device is in a reduced power state.

8.3 SLEEP Mode

8.3.1 Device Sleep

When the sleep conditions are met and the device is in REST (RELAX) mode, the device goes into SLEEP mode with periodic wake-ups to reduce power consumption. The device returns to NORMAL mode if any exit sleep condition is met.

Status	Condition	Action
Activate	SMBus low for BusTimeout ⁽¹⁾ AND DA Config[SLEEP] = 1 ⁽¹⁾ AND $ Current() \leq \text{Sleep Current}$ AND Voltage Time > 0 AND OperationStatus()[SDM] = 0 AND No PFAAlert() bits set AND No PFStatus() bits set AND No SafetyAlert() bits set AND No [AOLD], [AOLDL], [ASCD], [ASCDL] set in SafetyStatus()	Turn off CHG FET and PCHG FET if FET Options[SLEEPCHG] = 0 Device goes to sleep. Device wakes up every Sleep:Voltage Time period to measure voltage and temperature. Device wakes up every Sleep:Current Time period to measure current.
Exit	SMBus connected ⁽¹⁾ OR SMBus command received ⁽²⁾ OR DA Config[SLEEP] = 1 ⁽¹⁾ OR $ Current() > \text{Sleep Current}$ OR Voltage Time = 0 OR OperationStatus()[SDM] = 1 OR PFAAlert() bits set OR PFStatus() bits set OR SafetyAlert() bits set OR [AOLD], [AOLDL], [ASCD], [ASCDL] set in SafetyStatus()	Return to NORMAL mode

⁽¹⁾ **DA Config[SLEEP]** and SMBus low are not checked if the **ManufacturerAccess()** SLEEP mode command is used to enter SLEEP mode.

⁽²⁾ A wake on an SMBus command is only possible when the gas gauge is put to sleep using the **ManufacturerAccess()** SLEEP mode command. Otherwise, the gas gauge wakes on an SMBus connection (clock or data high).

The configuration options for SLEEP are in the following data flash.

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Power	Sleep	Sleep Current	I2	0	32767	10	mA	$ Current() $ threshold to enter SLEEP mode
Power	Sleep	Bus Timeout	U1	0	255	5	s	SMBus low time to enter SLEEP mode
Power	Sleep	Voltage Time	U1	0	255	5	s	Voltage sampling period in SLEEP mode
Power	Sleep	Current Time	U1	0	255	20	s	Current sampling period in SLEEP mode

8.3.2 ManufacturerAccess() MAC Sleep

The Sleep MAC command can override the requirement for bus low to enter SLEEP. In this case, the clock and data high condition are ignored for SLEEP to exit, though SLEEP will also exit if there is any further SMBus communication. The device can be sent to SLEEP with *ManufacturerAccess()* if specific sleep entry conditions are met.

8.3.3 IN SYSTEM SLEEP Mode

The BQ78350-R1A provides an option to enter SLEEP mode when the battery is in the system. When the **DA Configuration [IN_SYSTEM_SLEEP] = 1**, the device will enter SLEEP mode when *OperationStatus()[PRES] = 1* and all other sleep conditions are also met.

In the IN SYSTEM SLEEP mode, it is possible to read the data if **[IN_SYSTEM_SLEEP] = 1** and Bus Timeout = 0. This setting allows the gauge to enter SLEEP mode with active communication in progress.

8.4 SHUTDOWN Mode

8.4.1 Voltage Based Shutdown

To minimize power consumption and avoid draining the battery, the device can be configured to shutdown at a programmable voltage threshold. In SHUTDOWN mode, the device turns off the FETs after **FET Off Time**, and then shuts down to minimize power consumption after Delay time. Both **FET Off Time** and **Delay** time are referenced to the time the gauge receives the command. Thus, the **Delay** time must be set longer than the **FET Off Time**. When the device is in PERMANENT FAILURE mode, the parameters **PF Shutdown Voltage** and **PF Shutdown Time** configure the voltage-based shutdown.

Status	Condition	Action
Enable	Min(Cell Voltage in <i>DAStatus1()</i>) < Shutdown Voltage	<i>OperationStatus()[SDV]</i> = 1
Trip	Min(Cell Voltage in <i>CellVoltage1..15()</i>) continuous < Shutdown Voltage for Shutdown Time AND <i>Current()</i> <= 0	Turn DSG FET off for Shutdown Time
Shutdown	Protection Configuration:VAUXR = 0	Send device into SHUTDOWN mode
Shutdown	Protection Configuration:VAUXR = 1 AND <i>VAUXVoltage()</i> < Charger Present Threshold	Send device into SHUTDOWN mode
Exit	Voltage at TS1 pin > V _{BOOT}	<i>OperationStatus()[SDV]</i> = 0 Return to NORMAL mode

NOTE: The device goes through a full reset when exiting from SHUTDOWN mode, which means the device will re-initialize. The RAM data is re-loaded with a data flash setting. This is different than a partial reset, which could occur during a short power glitch. The device will check for the RAM integrity at partial reset, and if the data checksum is correct, RAM data will not be re-initialized.

The configuration options for SHUTDOWN are in the following data flash.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Power	Shutdown	Shutdown Voltage	I2	0	32767	1750	mV
Power	Shutdown	PF Shutdown Voltage	I2	0	32767	1750	mV
Power	Shutdown	Shutdown Time	U2	0	255	10	s
Power	Shutdown	PF Shutdown Time	U2	0	255	10	s
Power	Shutdown	Charger Present Threshold	I2	0	32767	3000	mV
Power	Ship	FET Off Time	U1	0	127	20	s
Power	Ship	Delay	U1	0	254	20	s
Power	Ship	Auto Ship Time	U2	0	65535	1440	min

8.4.2 Time Based Shutdown

The device can be configured to shut down after staying in SLEEP mode without communication for a preset time interval specified in **Auto Ship Time**. Setting the **PowerConfig[AUTO_SHIP_EN]** = 1 enables this feature. If **Auto Ship Time** is set to 0, this feature is disabled. Any communication to the device will restart the timer. When the timer reaches the **Auto Ship Time**, the time based shutdown effectively triggers the MAC shutdown command to start the shutdown sequence. The device returns to NORMAL mode when voltage at PACK pin > $V_{STARTUP}$.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Settings	Configuration	Power Config	H1	0x00	0x01	0x00	Hex

7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	AUTO_SHIP_EN

RSVD (Bits 7–1): Reserved. Do not use.

AUTO_SHIP_EN (Bit 0): Automatically shut down for shipment

- 1 = Enable auto shutdown after the device is in SLEEP mode without communication for a set period of time.
- 0 = Disable auto shutdown feature

8.4.3 ManufacturerAccess() MAC Shutdown

In SHUTDOWN mode, the device turns off the FETs after **FET Off Time**, and then shuts down to minimize power consumption after **Delay** time. Both **FET Off Time** and **Delay** time are referenced to the time the gauge receives the command. Thus, the **Delay** time must be set longer than the **FET Off Time**. The device returns to NORMAL mode when voltage at TS1 pin > V_{BOOT} . The device can be sent to this mode with the **ManufacturerAccess() Shutdown** command if **Current()** <= 0 and **OperationStatus()** [DSG] = 1.

8.5 Power Mode Indication (PWRM)

The PWRM pin can be used to indicate the power mode of the BQ78350-R1A. The PWRM has the following conditions:

- PWRM is High-Z (externally pulled high):
 - BQ78350-R1A is in NORMAL mode.
 - BQ78350-R1A is in SLEEP mode AND **SBS Gauging Configuration[PWRMSleep]** = 1.
 - Once in SHUTDOWN mode, the device has no control over PWRM.
- PWRM is Low:
 - BQ78350-R1A is in SLEEP mode AND **SBS Gauging Configuration[PWRMSleep]** = 0.
 - BQ78350-R1A prepares to enter SHUTDOWN mode.

This pin can be used to control other external circuit elements based on the power mode state of the BQ78350-R1A.

CEDV Gas Gauging

9.1 Introduction

The BQ78350-R1A features the Compensated End-of-Discharge Voltage (CEDV) gauging algorithm, capable of gauging a Li-Ion or LiFePO4 battery. The data from the gas gauge is in either mAh or mWh units based on the 0 or 1 setting of *[CapM]* in *BatteryMode()*, and can be scaled per the *[IPScale]* setting in *SpecInfo()*.

The operational overview in Figure 9-1 illustrates the gas gauge operation of the BQ78350-R1A.

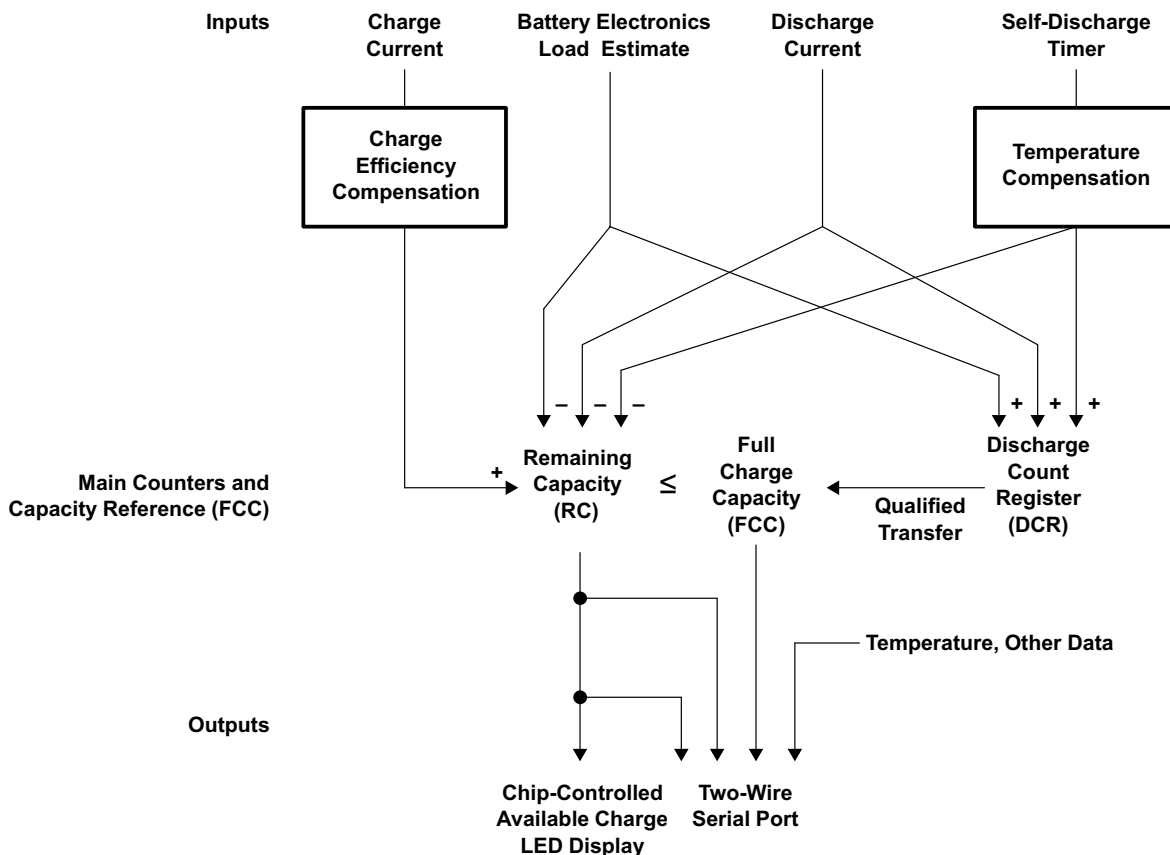


Figure 9-1. CEDV Operational Overview

The BQ78350-R1A accumulates the measured quantities of charge and discharge and estimates self-discharge of the battery. The BQ78350-R1A compensates the charge current measurement for temperature and state-of-charge of the battery. The BQ78350-R1A also adjusts the self-discharge estimation based on temperature.

9.1.1 Main Fuel Gauge Registers

The main charge counter, *RemainingCapacity()* (RC), represents the available capacity or energy in the battery at any given time. The BQ78350-R1A adjusts RC for charge, self-discharge, and other compensation factors. The information in the RC register is accessible through the SMBus.

The BQ78350-R1A computes RC in units based on the settings of two configuration bits, **CapM** and *SpecificationInfo()*. RC counts up during charge to a maximum value of FCC and down during discharge and self-discharge to a minimum of 0. In addition to charge and self-discharge compensation, the BQ78350-R1A calibrates RC at three low-battery-voltage thresholds, EDV2, EDV1, and EDV0. This provides a voltage-based calibration to the RC counter and is based on the lowest voltage measured at the BAT pin.

The Design Capacity (DC) register is the user-specified battery full capacity. It is calculated from **Design Capacity** and is represented in units set by **CapM**. It also represents the full-battery reference for the absolute display mode and *AbsoluteStateOfCharge()*. In programming **Design Capacity**, the value should not include the value programmed in **Reserve Capacity**.

The *FullChargeCapacity()* (FCC) register represents the initial or last measured full discharge of the battery. It is used as the battery full-charge reference for relative capacity indication. The BQ78350-R1A updates FCC after the battery undergoes a qualified discharge from nearly full to a low battery level. FCC is accessible through the SMBus.

The BQ78350-R1A computes FCC in units based on the settings of two configuration bits, **CapM** and **IPScale**. On initialization, the BQ78350-R1A sets FCC to the value stored in **Full Charge Capacity**. During subsequent discharges, the BQ78350-R1A updates FCC with the last measured discharge capacity of the battery. The last measured discharge of the battery is based on the value in the DCR register after a qualified discharge occurs. Once updated, the BQ78350-R1A writes the new FCC value to data flash in mAh, scaled per the setting of **IPScale**, to **Full Charge Capacity**.

NOTE: Care should be taken to ensure that the correct scaling is used to ensure that the **Full Charge Capacity** does not exceed 65535 of the units configured by the scaling. If **Full Charge Capacity** is calculated to be above 65535, then it will roll over creating potentially uncorrectable error in the gauging algorithm.

9.1.2 Fuel Gauge Operating Modes

During a gauging operation, different features and functions occur based on whether the battery is discharging, charging, or in a rest state.

Entry and exit of each mode is controlled by data flash parameters in the subclass *Fuel Gauging: Current Thresholds* section.

- In RELAX mode or DISCHARGE mode, the *[DSG]* flag in *GaugingStatus()* is set.
- CHARGE mode is entered when *Current* goes above **Chg Current Threshold**.
- CHARGE mode is exited and RELAX mode is entered when *Current* goes below **Quit Current** for a period of **Chg Relax Time**.
- DISCHARGE mode is entered when *Current* goes below **(-)Dsg Current Threshold**.
- DISCHARGE mode is exited and RELAX mode is entered when *Current* goes above **(-)Quit Current** threshold for a period of **Dsg Relax Time**.

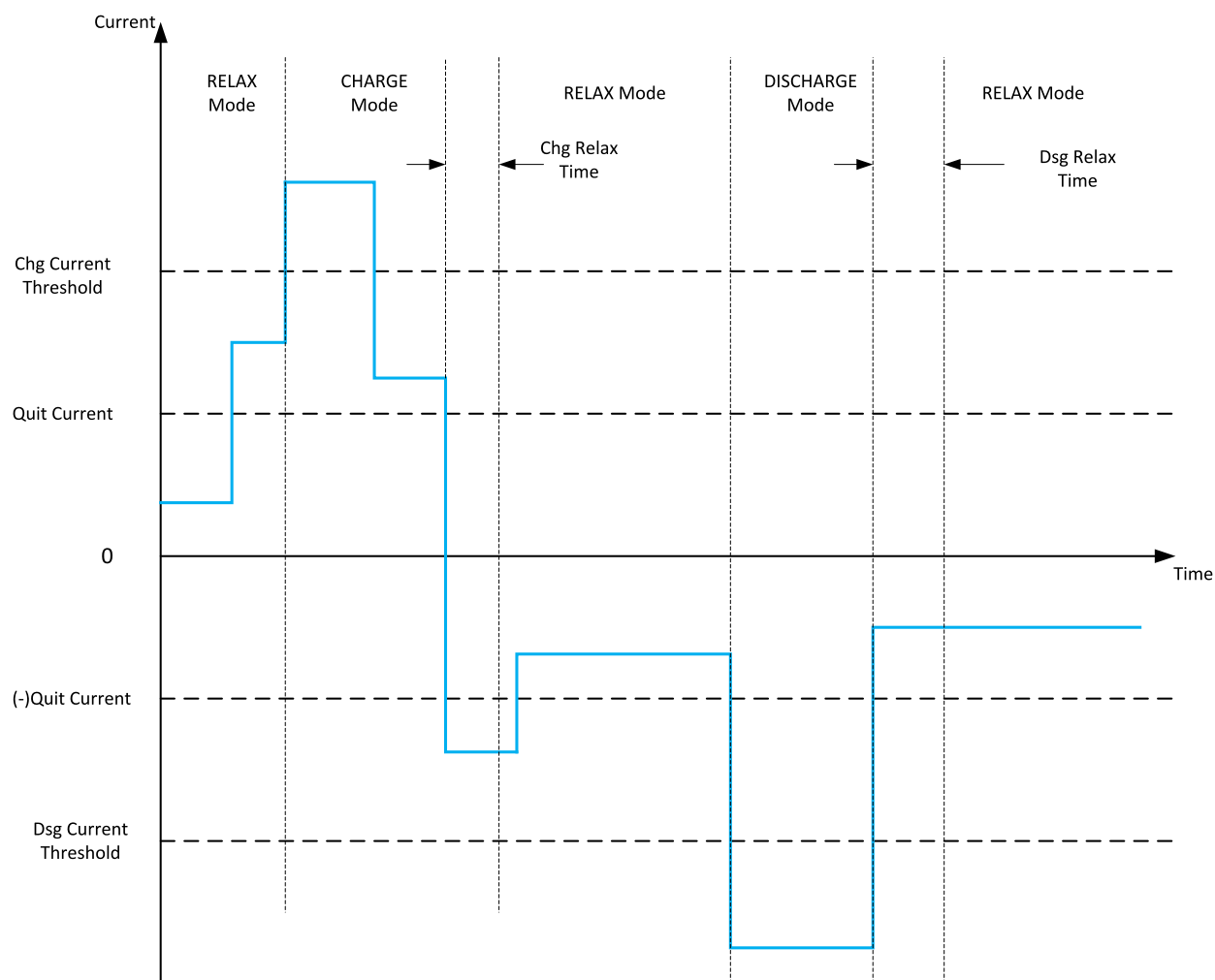


Figure 9-2. Fuel Gauge Operating Mode Example

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	Current Thresholds	Dsg Current Threshold	Unsigned Integer	2	0	2000	100	mA
Fuel Gauging	Current Thresholds	Dsg Relax Time	Unsigned Integer	1	0	255	1	s
Fuel Gauging	Current Thresholds	Chg Current Threshold	Unsigned Integer	2	0	2000	50	mA
Fuel Gauging	Current Thresholds	Chg Relax Time	Unsigned Integer	1	0	255	60	s
Fuel Gauging	Current Thresholds	Quit Current	Unsigned Integer	2	0	1000	10	mA

9.1.3 Full Charge Capacity

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	State	Learned Full Charge Capacity	Unsigned Integer	2	0	65535	4400	mAh or 10 mWh

FCC also represents the full battery reference for the relative display mode and *RelativeStateOfCharge()* calculations.

The *DischargeCountRegister()* (DCR) register that tracks discharge of the battery. The BQ78350-R1A uses the DCR register to update the FCC register if the battery undergoes a qualified discharge from nearly full to a low battery level. In this way, the BQ78350-R1A learns the true discharge capacity of the battery under system-use conditions.

The DCR counts up during discharge, independent of RC. DCR counts discharge activity, battery load estimation, and self-discharge increments. The BQ78350-R1A initializes DCR at the beginning of a discharge to FCC – RC when RC is within the programmed value in **Near Full**. The DCR initial value of FCC – RC is reduced by FCC/128 if **SC** = 1, and is not reduced if **SC** = 0. The DCR stops counting when the battery voltage reaches the EDV2 threshold on discharge.

9.1.4 Initial Battery Capacity at Device Reset

The BQ78350-R1A estimates the initial capacity of a battery pack at device reset, which is the case when battery cell(s) are first attached to the application circuit. The initial FCC is a direct copy of **Full Charge Capacity**. The initial RC and RSOC are estimated using the open-circuit voltage (OCV) characteristics of the programmed Li-Ion chemistry (default ID1210), **DOD at EDV2**, and **Learned Full Charge Capacity**. When assessing the RC vs. OCV correlation, the BQ78350-R1A uses the applicable *CellVoltage1..15()* data even if *ExtAveCellVoltage()* data is available. Upon the update of RC and RSOC based on the OCV data, the *BatteryMode()* [CF] flag will be cleared. This gives a reasonably accurate RSOC; however, battery capacity learning is required in order to determine the most accurate FCC, RC, and RSOC.

The determined value of remaining capacity can be further scaled, if needed, through the value of **RemCap Init Percent**. Upon a reset, the final value of *RemainingCapacity()* is initialized from the **RemCap Init Percent** value of the initial OCV correlated value. This value should be programmed to match the scaling configured by **SpecificationInfo**.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	RemCap Init Percent	Unsigned Integer	1	0	110	100	%

During battery capacity learning, **Full Charge Capacity** and **DOD at EDV2** will be learned and updated. **Full Charge Capacity** should be initialized to the **Design Capacity**. DOD at EDV2 should be initialized to $(1 - (\text{Battery Low \%} / 100)) \times 16384$.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	State	DOD at EDV2	Integer	2	0	16384	15232	—

9.1.5 Capacity Learning (FCC Update)

The BQ78350-R1A updates FCC with an amount based on the value in DCR if a qualified discharge occurs. The new value for FCC equals the DCR value plus the value of nearly full and low battery levels, as shown in the following equation:

$$\text{FCC (new)} = \text{DCR (final)} = \text{DCR (initial)} + \text{Measured Discharge to EDV2} + (\text{FCC} \times (\text{Battery Low \%} / 100))$$

The new value of FCC can be limited to not go above the **Design Capacity** value if **FCC_LIMIT** in **CEDV Gauging Configuration** is set.

NOTE: **Learned Full Charge Capacity** limits an update to a minimum of 100.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Battery Low %	Unsigned Integer	2	0	65535	700	0.01%
Fuel Gauging	State	Learned Full Charge Capacity	Integer	2	0	32767	4400	mAh

Battery Low % should be set to match a capacity value that corresponds to the first or highest voltage point, EDV2. It should be chosen where the capacity sensitivity to voltage is easily detectable. It is a non-measured portion of the overall **Learned Full Charge Capacity**. If the target **Battery Low %** is changed in the design, ensure that the initial value of **DOD at EDV2** is also adjusted accordingly.

9.1.6 Qualified Discharge

A qualified discharge occurs if the battery discharges from $RC \geq FCC - \text{Near Full}$ to the EDV2 voltage threshold with the following conditions:

- No valid charge activity occurs during the discharge period. A valid charge is defined as a charge of 10 mAh into the battery.
- No more than 256 mAh of self-discharge or battery load estimation occurs during the discharge period.
- The temperature does not drop below the low temperature thresholds programmed in **Low Temp** during the discharge period.
- The battery voltage reaches the EDV2 threshold during the discharge period and the voltage is greater than or equal to the EDV2 threshold minus 256 mV when the BQ78350-R1A detected EDV2.
 - When **CEDV Gauging Configuration [VFLT_EN]** is set, a filter is added to the EDV detection that is set by **CEDV Min Delta V** to improve false triggering under pulsed load activity. If the latest compensated EDV2 voltage changes by more than CEDV Min Delta V from the previously calculated value, then the previous one is not updated.
- $-Current()$ remains $< \text{Overload Current}$ when EDV2 is reached.
- No overload condition exists when EDV2 threshold is reached, or if RC has dropped to **Battery Low %** $\times FCC$.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Near Full	Integer	2	0	65535	200	mAh or 10 mWh
Fuel Gauging	CEDV Cfg	Learning Low Temp	Integer	2	-100	255	119	0.1°C
Fuel Gauging	CEDV Cfg	Min Delta V Filter	Integer	2	0	32767	10	mV

The BQ78350-R1A sets $[VDQ] = 1$ in **GaugingStatus()** when a qualified discharge begins. The BQ78350-R1A sets $[VDQ] = 0$ if any disqualifying condition occurs. One complication may arise regarding the state of $[VDQ]$ if **[CSYNC]** is set in **CEDV Gauging Configuration**. When **[CSYNC]** is enabled, RC is written to equal FCC on valid primary charge termination. This capacity synchronization is done even if the condition $RC \geq FCC - \text{Near Full}$ is NOT satisfied at charge termination.

FCC cannot be reduced by more than **FCC Learn Down** or increased by more than **FCC Learn Up** during any single update cycle. The BQ78350-R1A saves the new FCC value to the data flash within 4 s of being updated.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	FCC Learn Down	Integer	2	0	65535	256	mAh or 10 mWh
Fuel Gauging	CEDV Cfg	FCC Learn Up	Integer	2	0	65535	512	mAh or 10 mWh

9.1.7 End-of-Discharge Thresholds and Capacity Correction

The BQ78350-R1A monitors the battery for three low-voltage thresholds, EDV0, EDV1, and EDV2. The BQ78350-R1A uses the lowest, single-cell value from individual cell voltage measurements for EDV threshold comparison when **CEDV Gauging Configuration [EDV_EXT_CELL]** = 0. However, if this bit = 1, then the *ExternalCellVoltage()* is used.

With either Compensated or Fixed EDV configurations, the configured voltage to be used must be equal to or below the appropriate voltage for the corresponding **EDV2,1,0 Hold Time** to ensure correct detection under all load types. EDV1 Detection and its associated hold time do not begin until EDV2 has been detected and the EDV2 flag is set. Similarly, EDV0 detection does not begin before the EDV1 flag is set.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Fixed EDV0	Integer	2	0	32767	3031	mV
Fuel Gauging	CEDV Cfg	Fixed EDV1	Integer	2	0	32767	3385	mV
Fuel Gauging	CEDV Cfg	Fixed EDV2	Integer	2	0	32767	3501	mV
Fuel Gauging	CEDV Cfg	EDV0 Hold Time	Unsigned Integer	1	1	255	1	s
Fuel Gauging	CEDV Cfg	EDV1 Hold Time	Unsigned Integer	1	1	255	1	s
Fuel Gauging	CEDV Cfg	EDV2 Hold Time	Unsigned Integer	1	1	255	1	s

If the **[EDV_CMP]** bit in **CEDV Gauging Configuration** is set, automatic EDV compensation is enabled and the BQ78350-R1A computes the EDV0, EDV1, and EDV2 thresholds based on values stored in **CEDV Cfg** subclass of data flash and the battery's current discharge rate and temperature. However, if **[FIXED_EDV0]** bit in **CEDV Gauging Configuration** is set, then even if **[EDV_CMP]** = 1, then EDV0 is a fixed voltage value and is not compensated.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	EMF	Unsigned Integer	2	0	65535	3743	mV
Fuel Gauging	CEDV Cfg	EDV C0 Factor	Unsigned Integer	2	0	65535	149	—
Fuel Gauging	CEDV Cfg	EDV R0 Factor	Unsigned Integer	2	0	65535	867	—
Fuel Gauging	CEDV Cfg	EDV T0 Rate Factor	Unsigned Integer	2	0	65535	4030	—
Fuel Gauging	CEDV Cfg	EDV R1 Rate Factor	Unsigned Integer	2	0	65535	316	—
Fuel Gauging	CEDV Cfg	EDV TC Factor	Unsigned Integer	1	0	255	9	—
Fuel Gauging	CEDV Cfg	EDV a0 Age Factor	Unsigned Integer	1	0	255	0	—

The BQ78350-R1A disables EDV detection if the measured battery discharge current magnitude ($-Current()$) meets or exceeds the **Overload Current** threshold, which is scaled by IPSCALE. The BQ78350-R1A resumes EDV threshold detection after C drops below the **Overload Current** threshold. Any EDV threshold detected is reset after charge is applied and [VDQ] is then cleared after RC has increased by a value of 10, which is scaled by IPSCALE.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Overload Current	Integer	2	0	32767	5000	mA

The BQ78350-R1A uses the EDV thresholds to apply voltage-based corrections to the RC register according to the content in [Table 9-1](#).

Table 9-1. State-of-Charge Based on Low Battery Voltage

Threshold	Relative State-of-Charge (RSOC)
EDV0	0%
EDV1	3%
EDV2	Battery Low %

The BQ78350-R1A performs EDV-based RC adjustments with $Current() \geq 3C/32$. No EDVs are set if $Current() < 3C/32$. The BQ78350-R1A adjusts RC as it detects each threshold. If the voltage threshold is reached before the corresponding capacity on discharge, the BQ78350-R1A reduces RC to the appropriate amount, as shown in [Table 9-1](#).

If an RC % level is reached on discharge before the voltage reaches the corresponding threshold, then RC is held at that % level until the threshold is reached. RC is only held if [VDQ] = 1, indicating a valid learning cycle is in progress. If **Battery Low %** is set to 0, EDV1 and EDV0 corrections are disabled.

9.1.8 Reserve Capacity

The BQ78350-R1A can provide an additional programmable quantity of capacity in "reserve"; that is, when $RC = 0$, then there is still **Reserve Capacity** left. This value is required to be entered and scaled to match the settings of IPSCALE and CapM settings.

The value of **Reserve Capacity** is subtracted from the learn capacity when determining the value of the reported FCC. This means when $RSOC = 0\%$ (EDV0), then there is still some capacity left for critical system actions. It is strongly recommended that when determining the value for Reserve Capacity that the setting of **Battery Low %** is still considered to ensure the appropriate setting of the EDV2 voltage on the discharge curve. For example: If Reserve Capacity ~1% of **Design Capacity**, then the typical value for **Battery Low %** would be 6%.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Reserve Capacity	Integer	2	0	32767	0	mAh

9.1.9 EDV Discharge Rate and Temperature Compensation

If EDV compensation is enabled, $CEDV = 1$, the BQ78350-R1A calculates battery voltage to determine EDV0, EDV1, and EDV2 thresholds as a function of battery capacity, temperature, and discharge load.

The general equation for EDV0, EDV1, and EDV2 calculation is as follows:

$$EDV_{0,1,2} = n (EMF \times FBL - |ILOAD| \times R0 \times FTZ)$$

- EMF is a no-load cell voltage higher than the highest cell EDV threshold computed. EMF is programmed in mV in **EMF**.
- ILOAD is the current discharge load magnitude.
- n = the number of series cells. In the BQ78350-R1A case $n = 1$.

- FBL is the factor that adjusts the EDV voltage for battery capacity and temperature to match the no-load characteristics of the battery.

$$FBL = f(C0, C + C1, T)$$

- C (either 0%, 3%, or **Battery Low %** for EDV0, EDV1, and EDV2, respectively) and C0 are the capacity related EDV adjustment factors. C0 is programmed in **EDV C0 Factor**. C1 is the desired residual battery capacity remaining at EDV0 (RC = 0). The C1 factor is stored in **EDV C1 Factor**.
- T is the current temperature in K.
- R0•FTZ represents the resistance of a cell as a function of temperature and capacity.

$$FTZ = f(R1, T0, C + C1, TC)$$

- R0 is the first order rate dependency factor stored in **EDV R0 Factor** (DF).
- T is the current temperature. C is the battery capacity relating to EDV0, EDV1, and EDV2.
- R1 adjusts the variation of impedance with battery capacity. R1 is programmed in **EDV R1 Rate Factor**.
- T0 adjusts the variation of impedance with battery temperature. T0 is programmed in **EDV T0 Rate Factor**.
- TC adjusts the variation of impedance for cold temperatures $T < TC$ is programmed in **EDV TC Factor**.
- Typical values for the EDV compensation factors, based on overall pack voltages for a 3s2p Li-Ion 18650 pack, are
 - EMF = 11550/3
 - T0 = 4475
 - C0 = 235
 - C1 = 0
 - R0 = 5350/3
 - R1 = 250
 - TC = 3

The graphs below show the calculated EDV0, EDV1, and EDV2 thresholds versus capacity using the typical compensation values for different temperatures and loads for a Li-Ion 18650 cell. The compensation values vary widely for different cell types and manufacturers and must be matched exactly to the unique characteristics for optimal performance.

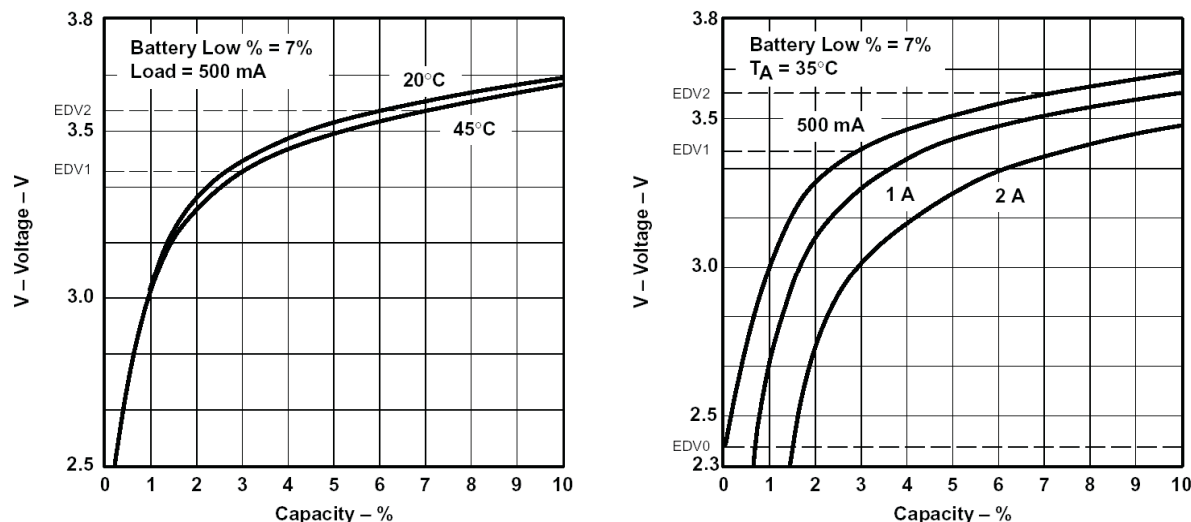


Figure 9-3. (a) EDV Calculations vs Various Temperatures, (b) EDV Calculations vs Capacity for Various Loads

9.1.10 EDV Age Factor

EDV Age factor allows the BQ78350-R1A to correct the EDV detection algorithm to compensate for cell aging. This parameter scales cell impedances as the cycle count increases. This factor is used to accommodate for much higher impedances observed in larger capacity and/or aged cells.

For most Li-Ion and Li-Polymer applications, the default value of zero is sufficient. However, for Lithium Iron Phosphate, a value of 18 is recommended.

9.1.11 Self Discharge

The BQ78350-R1A estimates the self-discharge of the battery to maintain an accurate measurement of the battery capacity during periods of inactivity. The BQ78350-R1A makes self-discharge adjustments to RC every ¼ second when awake and periodically when in SLEEP mode. The period is determined by **Sleep Current Time**.

The self-discharge estimation rate for 25°C is doubled for each 10 degrees above 25°C or halved for each 10 degrees below 25°C. The table below shows the relation of the self-discharge estimation at a given temperature to the rate programmed for 25°C.

Temperature (°C)	Self-Discharge Rate (Average)
Temp < 10	¼ Y% per day
10 ≤ Temp < 20	½ Y% per day
20 ≤ Temp < 30	1 Y% per day
30 ≤ Temp < 40	2 Y% per day
40 ≤ Temp < 50	4 Y% per day
50 ≤ Temp < 60	8 Y% per day
60 ≤ Temp < 70	16 Y% per day
70 ≤ Temp	32 Y% per day

The nominal self-discharge rate, %PERDAY (% per day), is programmed in an 8-bit value **Self-Discharge Rate** by the following relation:

$$\text{Self-Discharge Rate} = \%PERDAY / 0.01$$

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Self Discharge Rate	Unsigned Integer	1	0%	255%	20%	0.01/day

9.1.12 Battery Electronic Load Compensation

The BQ78350-R1A can be configured to compensate for a constant load (as from battery electronics) present in the battery pack at all times. The BQ78350-R1A applies the compensation continuously when the charge or discharge is below the digital filter. The BQ78350-R1A applies the compensation in addition to self-discharge.

The compensation occurs at a rate determined by the value stored in **Electronics Load**. The compensation range is 0 µA – 765 µA in steps of approximately 3 µA.

The amount of internal battery electronics load estimate in µA, BEL, is stored as follows: **Electronics Load** = BEL/3.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Fuel Gauging	CEDV Cfg	Electronics Load	Integer	2	0	255	0	3 µA

9.2 Gauging Configuration Options

The BQ78350-R1A has a variety of configurable options that can be configured, enabled, or disabled through the following data flash options.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Gas Gauging	Design	Design Capacity mAh	I2	0	32767	4400	mAh
Gas Gauging	Design	Design Capacity cWh	I2	0	32767	6336	cWh
Gas Gauging	Design	Design Voltage	I2	0	5000	3600	mV
Gas Gauging	Cycle	Cycle Count Percentage	U1	0%	100%	90%	

Table 9-2. SBS Gauging Configuration

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	SBS Gauging Configuration	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
PWRMSleep	RSVD	RSVD	RSVD	RSVD	RSVD	RSOC_HOLD	RSOCL

PWRMSleep (Bit 7): Power mode indication pin control during sleep

1 = PWRM is high during sleep.

0 = PWRM is low during sleep (default).

RSOC_HOLD (Bit 1): Prevent RSOC from increasing during discharge

1 = RSOC not allowed to increase during discharge

0 = RSOC not limited (default)

RSOCL (Bit 0): *RelativeStateOfCharge()* and *RemainingCapacity()* behaviors at end of charge

1 = Held at 99% until valid charge termination. On entering valid charge termination, updates to 100%.

0 = Actual Value Shown (Default)

Table 9-3. CEDV Gauging Configuration

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	CEDV Gauging Configuration	Hex	2	0x0000	0xFFFF	0x0002	—

15	14	13	12	11	10	9	8
RSVD	RSVD	RSVD	RSVD	RSVD	FC_FOR_VDQ	RSVD	FCC_LIMIT
7	6	5	4	3	2	1	0
VFLT_EN	RSVD	FIXED_EDV0	SC	EDV_CMP	EDV_EXT_CELL	CSYNC	CCT

RSVD (Bits 15–11): Reserved. Do not use.

FC_FOR_VDQ (Bit 10): Set VDQ on setting of FC

1 = VDQ set on setting of FC

0 = VDQ set on start of qualified discharge (default)

RSVD (Bit 9): Reserved. Do not use.

FCC_LIMIT (Bit 8): Prevent FCC from learning higher than Design Capacity

- 1 = Enabled
- 0 = Disabled (default)

VFLT_EN (Bit 7): Enable voltage filtering to prevent sudden termination due to pulse loading

- 1 = Enabled
- 0 = Disabled (default)

RSVD (Bit 6): Reserved. Do not use.

FIXED_EDV0 (Bit 5): This bit determines whether the BQ78350-R1A implements automatic EDV compensation to calculate the EDV0 threshold based on rate, temperature, and capacity, or uses a fixed voltage value. If EDV_CMP = 0, then this bit has no effect.

- 1 = EDV Compensation Not Used. For example: Fixed EDV gauge enabled
- 0 = EDV Compensation Used (default)

SC (Bit 4): This bit enables learning cycle optimization for a Smart Charger or independent charge.

- 1 = Learning cycle optimized for independent charger
- 0 = Learning cycle optimized for Smart Charger (default)

EDV_CMP (Bit 3): This bit enables EDV Compensation for EDV2, EDV1, and EDV0.

- 1 = Enabled
- 0 = Disabled (default)

EDV_EXT_CELL (Bit 2): External average cell voltage used for EDV detection

- 1 = External average cell voltage used as EDV detection reference
- 0 = Minimum individual cell voltage used as EDV detection reference (default)

CSYNC (Bit 1): Sync *RemainingCapacity()* with *FullChargeCapacity()* at valid charge termination

- 1 = Synchronized (default)
- 0 = Not synchronized

CCT (Bit 0): Cycle count threshold

- 1 = Use CC % of *FullChargeCapacity()*
- 0 = Use CC % of DesignCapacity (default)

Lifetime Data Collection

10.1 Description

The device has extensive capabilities for logging events over the life of the battery useful for analysis. The Lifetime Data Collection is enabled by setting **ManufacturingStatus[LF_EN]** = 1. The data is collected in RAM and only written to DF under the following conditions to avoid wear out of the data flash:

- Every 10 hours if RAM content is different from flash.
- In permanent fail, before data flash updates are disabled.
- Before scheduled shutdown
- Before low voltage shutdown

The lifetime data stops collecting under following conditions:

- After permanent fail
- Lifetime Data Collection is disabled by setting **ManufacturingStatus[LF_EN]** = 0.

Total firmware Runtime starts when lifetime data is enabled.

- Voltage
 - Max/Min Cell Voltage Each Cell
 - Max Delta Cell Voltage at any given time (that is, the max cell imbalance voltage)
- Current
 - Max Charge/Discharge Current
 - Max Average Discharge Current
 - Max Average Discharge Power
- Safety Events that trigger the *SafetyStatus()* (The 12 most common are tracked.)
 - Number of Safety Events
 - Cycle Count at Last Safety Event(s)
- Charging Events
 - Number of Valid Charge Terminations (That is, the number of times [VCT] is set.)
 - Cycle Count at Last Charge Termination
- Gauging Events
 - Cycle Count at Last FCC update
- Power Events
 - Number of shutdowns
- Cell Balancing (This data is updated every two hours.)
 - Cell Balancing Time each Cell
- Temperature
 - Max/Min Cell Temp
 - Delta Cell Temp (max delta cell temperature across the thermistors that are used to report cell temperature)
- Time (This data is updated every 2 hours.)
 - Total runtime
 - Time spent different temperature ranges (See [Charge Algorithm](#) for ranges.)

10.2 Lifetimes

The Lifetime Data commands 0x60 through 0x66 provide an array of historical data for diagnostic analysis. The data is listed in byte order from LSB to MSB.

10.2.1 *LifetimeDataBlock1()* 0x0060

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Voltage	Max Voltage Cell 1	U2	0	32767	0	mV	Maximum reported cell voltage 1
Lifetimes	Voltage	Max Voltage Cell 2	U2	0	32767	0	mV	Maximum reported cell voltage 2
Lifetimes	Voltage	Max Voltage Cell 3	U2	0	32767	0	mV	Maximum reported cell voltage 3
Lifetimes	Voltage	Max Voltage Cell 4	U2	0	32767	0	mV	Maximum reported cell voltage 4
Lifetimes	Voltage	Max Voltage Cell 5	U2	0	32767	0	mV	Maximum reported cell voltage 5
Lifetimes	Voltage	Max Voltage Cell 6	U2	0	32767	0	mV	Maximum reported cell voltage 6
Lifetimes	Voltage	Max Voltage Cell 7	U2	0	32767	0	mV	Maximum reported cell voltage 7
Lifetimes	Voltage	Max Voltage Cell 8	U2	0	32767	0	mV	Maximum reported cell voltage 8
Lifetimes	Voltage	Max Voltage Cell 9	U2	0	32767	0	mV	Maximum reported cell voltage 9
Lifetimes	Voltage	Max Voltage Cell 10	U2	0	32767	0	mV	Maximum reported cell voltage 10
Lifetimes	Voltage	Max Voltage Cell 11	U2	0	32767	0	mV	Maximum reported cell voltage 11
Lifetimes	Voltage	Max Voltage Cell 12	U2	0	32767	0	mV	Maximum reported cell voltage 12
Lifetimes	Voltage	Max Voltage Cell 13	U2	0	32767	0	mV	Maximum reported cell voltage 13
Lifetimes	Voltage	Max Voltage Cell 14	U2	0	32767	0	mV	Maximum reported cell voltage 14
Lifetimes	Voltage	Max Voltage Cell 15	U2	0	32767	0	mV	Maximum reported cell voltage 15

10.2.2 *LifetimeDataBlock2()* 0x0061

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Voltage	Min Voltage Cell 1	U2	0	32767	255	mV	Minimum reported cell voltage 1
Lifetimes	Voltage	Min Voltage Cell 2	U2	0	32767	0	mV	Minimum reported cell voltage 2
Lifetimes	Voltage	Min Voltage Cell 3	U2	0	32767	0	mV	Minimum reported cell voltage 3
Lifetimes	Voltage	Min Voltage Cell 4	U2	0	32767	0	mV	Minimum reported cell voltage 4
Lifetimes	Voltage	Min Voltage Cell 5	U2	0	32767	0	mV	Minimum reported cell voltage 5
Lifetimes	Voltage	Min Voltage Cell 6	U2	0	32767	0	mV	Minimum reported cell voltage 6
Lifetimes	Voltage	Min Voltage Cell 7	U2	0	32767	0	mV	Minimum reported cell voltage 7
Lifetimes	Voltage	Min Voltage Cell 8	U2	0	32767	0	mV	Minimum reported cell voltage 8
Lifetimes	Voltage	Min Voltage Cell 9	U2	0	32767	0	mV	Minimum reported cell voltage 9
Lifetimes	Voltage	Min Voltage Cell 10	U2	0	32767	0	mV	Minimum reported cell voltage 10
Lifetimes	Voltage	Min Voltage Cell 11	U2	0	32767	0	mV	Minimum reported cell voltage 11
Lifetimes	Voltage	Min Voltage Cell 12	U2	0	32767	0	mV	Minimum reported cell voltage 12
Lifetimes	Voltage	Min Voltage Cell 13	U2	0	32767	0	mV	Minimum reported cell voltage 13
Lifetimes	Voltage	Min Voltage Cell 14	U2	0	32767	0	mV	Minimum reported cell voltage 14
Lifetimes	Voltage	Min Voltage Cell 15	U2	0	32767	0	mV	Minimum reported cell voltage 15

10.2.3 *LifetimeDataBlock3()* 0x0062

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Voltage	Max Delta Cell Voltage	I2	0	32767	0	mV	Maximum reported delta between cell voltages 1 to 15
Lifetimes	Current	Max Chg Current	I2	0	32767	0	mA	Maximum reported <i>Current()</i> in charge direction
Lifetimes	Current	Max Dsg Current	I2	0	32767	0	mA	Maximum reported <i>Current()</i> in discharge direction
Lifetimes	Current	Max Avg Dsg Current	I2	0	32767	0	mA	Maximum reported <i>AverageCurrent()</i> in discharge direction

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Current	Max Avg Dsg Power	I2	0	32767	0	cW	Maximum reported Power in discharge direction
Lifetimes	Temperature	Max Cell Temp	I1	–128	127	–128	°C	Maximum reported cell temperature
Lifetimes	Temperature	Min Cell Temp	I1	–128	127	127	°C	Minimum reported cell temperature
Lifetimes	Temperature	Max Delta Temp Cell	I1	–128	127	0	°C	Maximum reported temperature delta for TSx inputs configured as cell temperature
Lifetimes	Temperature	Max FET Temp	I1	–128	127	–128	°C	Maximum reported FET temperature

10.2.4 LifetimeDataBlock4() 0x0063

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Power Events	No of Shutdowns	U1	0	255	0	events	Total number of Shutdown events
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Cell Balancing	CB Time Cell 1	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 1
Lifetimes	Cell Balancing	CB Time Cell 2	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 2
Lifetimes	Cell Balancing	CB Time Cell 3	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 3
Lifetimes	Cell Balancing	CB Time Cell 4	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 4
Lifetimes	Cell Balancing	CB Time Cell 5	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 5
Lifetimes	Cell Balancing	CB Time Cell 6	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 6
Lifetimes	Cell Balancing	CB Time Cell 7	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 7
Lifetimes	Cell Balancing	CB Time Cell 8	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 8
Lifetimes	Cell Balancing	CB Time Cell 9	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 9
Lifetimes	Cell Balancing	CB Time Cell 10	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 10
Lifetimes	Cell Balancing	CB Time Cell 11	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 11
Lifetimes	Cell Balancing	CB Time Cell 12	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 12
Lifetimes	Cell Balancing	CB Time Cell 13	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 13
Lifetimes	Cell Balancing	CB Time Cell 14	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 14
Lifetimes	Cell Balancing	CB Time Cell 15	U1	0	255	0	2 h	Total performed cell balancing bypass time cell 15

10.2.5 LifetimeDataBlock5() 0x0064

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Time	Total Fw Runtime	U2	0	65535	0	2 h	Total firmware runtime between resets
Lifetimes	Time	Time Spent in UT	U2	0	65535	0	2 h	Total firmware runtime spent below T1
Lifetimes	Time	Time Spent in LT	U2	0	65535	0	2 h	Total firmware runtime spent between T1 and T2
Lifetimes	Time	Time Spent in ST	U2	0	65535	0	2 h	Total firmware runtime spent between T2 and T3
Lifetimes	Time	Time Spent in HT	U2	0	65535	0	2 h	Total firmware runtime spent between T3 and T4
Lifetimes	Time	Time Spent in OT	U2	0	65535	0	2 h	Total firmware runtime spent above T6
Lifetimes	Time	Time Since Last Charge	U2	0	65535	0	2 h	Total firmware runtime since last valid charge termination

10.2.6 LifetimeDataBlock6() 0x0065

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Safety Events	No of COV Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [COV] events
Lifetimes	Safety Events	Last COV Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [COV] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of CUV Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [CUV] events
Lifetimes	Safety Events	Last CUV Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [CUV] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of OCD Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OCD] events
Lifetimes	Safety Events	Last OCD Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [OCD] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of OCC Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OCC] events
Lifetimes	Safety Events	Last OCC Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [OCC] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of AOLD Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OLD] events
Lifetimes	Safety Events	Last AOLD Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [OLD] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of ASCD Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [SCD] events
Lifetimes	Safety Events	Last ASCD Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [SCD] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of OTC Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OTC] events
Lifetimes	Safety Events	Last OTC Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [OTC] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of OTD Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OTD] events
Lifetimes	Safety Events	Last OTD Event	U2	0	32767	0	cycles	Last <i>SafetyStatus()</i> [OTD] event in <i>CycleCount()</i> cycles

10.2.7 LifetimeDataBlock7() 0x0066

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Safety Events	No of OTF Events	U2	0	32767	0	events	Total number of <i>SafetyStatus()</i> [OTF] events
Lifetimes	Safety Events	Last OTF Event	U2	0	32767	0	events	Last <i>SafetyStatus()</i> [OTF] event in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of Valid Charge Terminations	U2	0	32767	0	events	Total number of valid charge termination events

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Lifetimes	Safety Events	Last Valid Charge Termination	U2	0	32767	0	cycles	Last valid charge termination in <i>CycleCount()</i> cycles
Lifetimes	Safety Events	No of FCC Update Events	U2	0	32767	0	events	Total number of FCC update events
Lifetimes	Safety Events	Last FCC Update	U2	0	32767	0	cycles	Last FCC update event in <i>CycleCount()</i> cycles
Lifetimes	Power Events	No of Shutdowns	U1	0	255	0	events	Total number of Shutdown events
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Power Events	Reserved	U1	0	255	0	—	Reserved
Lifetimes	Cell Balancing	CB Time Cell 1	U1	0	255	0	2h	Total performed cell balancing bypass time cell 1
Lifetimes	Cell Balancing	CB Time Cell 2	U1	0	255	0	2h	Total performed cell balancing bypass time cell 2
Lifetimes	Cell Balancing	CB Time Cell 3	U1	0	255	0	2h	Total performed cell balancing bypass time cell 3
Lifetimes	Cell Balancing	CB Time Cell 4	U1	0	255	0	2h	Total performed cell balancing bypass time cell 4
Lifetimes	Cell Balancing	CB Time Cell 5	U1	0	255	0	2h	Total performed cell balancing bypass time cell 5
Lifetimes	Cell Balancing	CB Time Cell 6	U1	0	255	0	2h	Total performed cell balancing bypass time cell 6
Lifetimes	Cell Balancing	CB Time Cell 7	U1	0	255	0	2h	Total performed cell balancing bypass time cell 7
Lifetimes	Cell Balancing	CB Time Cell 8	U1	0	255	0	2h	Total performed cell balancing bypass time cell 8
Lifetimes	Cell Balancing	CB Time Cell 9	U1	0	255	0	2h	Total performed cell balancing bypass time cell 9
Lifetimes	Cell Balancing	CB Time Cell 10	U1	0	255	0	2h	Total performed cell balancing bypass time cell 10
Lifetimes	Cell Balancing	CB Time Cell 11	U1	0	255	0	2h	Total performed cell balancing bypass time cell 11
Lifetimes	Cell Balancing	CB Time Cell 12	U1	0	255	0	2h	Total performed cell balancing bypass time cell 12
Lifetimes	Cell Balancing	CB Time Cell 13	U1	0	255	0	2h	Total performed cell balancing bypass time cell 13
Lifetimes	Cell Balancing	CB Time Cell 14	U1	0	255	0	2h	Total performed cell balancing bypass time cell 14
Lifetimes	Cell Balancing	CB Time Cell 15	U1	0	255	0	2h	Total performed cell balancing bypass time cell 15

Device Security

11.1 Description

There are three levels of secured operation within the device. To switch between the levels, different operations are needed with different keys. The three levels are SEALED (SE), UNSEALED (UN), and FULL ACCESS (FA). The device also supports SHA-1 HMAC authentication with the host system.

11.2 SHA-1 Description

As of March 2012, the latest revision is FIPS 180–4. SHA-1, or secure hash algorithm, is used to compute a condensed representation of a message or data also known as hash. For messages $< 2^{64}$, the SHA-1 algorithm produces a 160-bit output called a digest.

In a SHA-1 one-way hash function, there is no known mathematical method of computing the input given, only the output. The specification of SHA-1, as defined by FIPS 180–4, states that the input consists of 512-bit blocks with a total input length less than 264 bits. Inputs that do not conform to integer multiples of 512-bit blocks are padded before any block is input to the hash function. The SHA-1 algorithm outputs the 160-bit digest.

The device generates a SHA-1 input block of 288 bits (total input = 160-bit message + 128-bit key). To complete the 512-bit block size requirement of the SHA-1 function, the device pads the key and message with a 1, followed by 159 0s, followed by the 64 bit value for 288 (000...00100100000), which conforms to the pad requirements specified by FIPS 180–4.

Detailed information about the SHA-1 algorithm can be found here:

1. <http://www.nist.gov/itl/>
2. <http://csrc.nist.gov/publications/fips>
3. www.faqs.org/rfcs/rfc3174.html

11.3 HMAC Description

The SHA-1 engine calculates a modified HMAC value. Using a public message and a secret key, the HMAC output is considered to be a secure fingerprint that authenticates the device used to generate the HMAC.

To compute the HMAC: Let H designate the SHA-1 hash function, M designate the message transmitted to the device, and KD designate the unique 128-bit Unseal/Full Access/Authentication key of the device. HMAC(M) is defined as:

$H[KD \parallel H(KD \parallel M)]$, where $\parallel$ symbolizes an append operation.

The message, M, is appended to the unseal/full access/authentication key, KD, and padded to become the input to the SHA-1 hash. The output of this first calculation is then appended to the unseal/full access/authentication key, KD, padded again, and cycled through the SHA-1 hash a second time. The output is the HMAC digest value.

11.4 Authentication

1. Generate 160-bit message M using a random number generator that meets approved random number generators described in FIPS PUB 140–2.
2. Generate SHA-1 input block B1 of 512 bytes (total input = 128-bit authentication key KD + 160-bit message M + 1 + 159 0s + 100100000).
3. Generate SHA-1 hash HMAC1 using B1.

4. Generate SHA-1 input block B2 of 512 bytes (total input = 128-bit authentication key KD + 160-bit hash HMAC1 + 1 + 159 0s + 100100000).
5. Generate SHA-1 hash HMAC2 using B2.
6. With no active *ManufacturerInput()* data waiting, write 160-bit message M to *ManufacturerInput()* in the format 0xAABBCCDDEEFFGGHHIIJJKLLMMNNOOPPQQRRSSTT, where AA is LSB.
7. Wait 250 ms, then read *ManufacturerInput()* for HMAC3.
8. Compare host HMAC2 with device HMAC3, and if it matches, both host and device have the same key KD and the device is authenticated.

11.5 Security Modes

11.5.1 FULL ACCESS or UNSEALED to SEALED

The *Seal Device* command instructs the device to limit access to the SBS functions and data flash space and sets the *[SEC1]/[SEC0]* flags. In SEALED mode, standard SBS functions have access per the Smart Battery Data Specification. Extended SBS functions and data flash are not accessible. Once in SEALED mode, the part can never permanently return to UNSEALED or FULL ACCESS modes although there is a capability to temporarily switch from SEALED to UNSEALED and then to FULL ACCESS.

11.5.2 SEALED to UNSEALED

SEALED to UNSEALED instructs the device to temporarily extend access to the SBS and data flash space and clears the *[SEC1]/[SEC0]* flags. In UNSEALED mode, all data, SBS, and DF have read/write access. Unsealing is a two-step command performed by writing the first word of the unseal key to *ManufacturerAccess()* (MAC), followed by the second word of the unseal key to *ManufacturerAccess()*. The unseal key can be read and changed via the *MAC SecurityKey()* command when in the FULL ACCESS mode. To return to the SEALED mode, either a hardware reset is needed, or the *MAC Seal Device()* command is needed to transit from FULL ACCESS or UNSEALED to SEALED.

11.5.3 UNSEALED to FULL ACCESS

UNSEALED to FULL ACCESS instructs the device to temporarily allow full access to all SBS commands and data flash. The device is shipped from TI in this mode. The keys for UNSEALED to FULL ACCESS can be read and changed via the MAC command *SecurityKey()* when in FULL ACCESS mode. Changing from UNSEALED to FULL ACCESS is performed by using the *ManufacturerAccess()* command, by writing the first word of the Full Access Key to *ManufacturerAccess()*, followed by the second word of the Full Access Key to *ManufacturerAccess()*. In FULL ACCESS mode, the command to go to boot ROM can be sent.

Manufacture Production

12.1 Manufacture Testing

To improve the manufacture testing flow, the BQ78350-R1A allows certain features to be toggled on or off through *ManufacturerAccess()* commands: for example, the *PRE-CHG FET()*, *CHG FET()*, *DSG FET()*, *Lifetime Data Collection()*, *Calibration()*, and so on. Enabling only the feature under test can simplify the test flow in production by avoiding any feature interference. These toggling commands will only set the RAM data, which means the conditions set by these commands will be cleared if a reset or seal is issued to the gauge. The *ManufacturingStatus()* keeps track of the status (enabled or disabled) of each feature.

The data flash **Mfg Status Init** provides the option to enable or disable individual features for normal operation. Upon a reset or a seal command, the *ManufacturingStatus()* will be re-loaded from data flash **ManufacturingStatus**. This also means if an update is made to *ManufacturingStatus()* to enable or disable a feature, the gauge will only take the new setting if a reset or seal command is sent.

12.1.1 Manufacturing Status Configuration

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Manufacturing	Mfg Status Init	Hex	2	0x0000	0xFFFF	0x00	—

15	14	13	12	11	10	9	8
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	LED_EN	SAFE_EN
7	6	5	4	3	2	1	0
BBR_EN	PF_EN	LF_EN	FET_EN	RSVD	RSVD	RSVD	RSVD

RSVD (Bits 15–10): Reserved. Do not use.

LED_EN (Bit 9): Enables $\overline{\text{DISP}}$ pin triggered activation of the display

1 = Enabled

0 = Disabled (default)

SAFE_EN (Bit 8): SAFE Action

1 = Enabled

0 = Disabled (default)

BBR_EN (Bit 7): Black Box Recorder

1 = Enabled

0 = Disabled (default)

PF_EN (Bit 6): Permanent Fail. The [DFW], [IFC], [SOV], and [SUV] permanent failure features are always enabled regardless of the setting of this bit.

1 = Enabled

0 = Disabled (default)

LF_EN (Bit 5): Lifetime Data Collection

- 1 = Enabled
- 0 = Disabled (default)

FET_EN (Bit 4): FET action

- 1 = Enabled
- 0 = Disabled (default)

RSVD (Bits 3–0): Reserved. Do not use.

12.2 Calibration

The BQ78350-R1A device has integrated routines that support calibration of current, voltage, and temperature readings, accessible after writing 0xF081 or 0xF082 to *ManufacturerAccess()* when the *ManufacturingStatus()[CAL]* bit is ON. While the calibration is active, the factory calibrated ADC data is available on *ManufacturerData()*. The device stops reporting calibration data on *ManufacturerData()* if any other MAC commands are sent or the device is reset or sealed.

NOTE: The *ManufacturingStatus()[CAL]* bit must be turned OFF after calibration is completed. This bit is cleared at reset or after sealing.

ManufacturerAccess()	Description
0x002D	Enables/Disables <i>ManufacturingStatus()[CAL]</i>
0xF080	Disables raw ADC data output on <i>ManufacturerData()</i>
0xF081	Outputs factory calibrated ADC data of the first 14-series cell voltages on <i>ManufacturerData()</i>
0xF082	Outputs factory calibrated ADC data of Cell Voltage 15, external average voltage, VAUX voltage, current, and temperatures on <i>ManufacturerData()</i>

For 0xF081, the *ManufacturerData()* output format is:

ZZYYaaAAabbBBccCCddDDeeEEffFFggGGhhHHiiJJjJkkKKllLLmmMMnnNNooOO, where:

Value	Format	Description
ZZ	Byte	8-bit counter, increments when raw ADC values are refreshed (every 250 ms)
YY	Byte	Output Status <i>ManufacturerAccess()</i> = 0xF081: 1 <i>ManufacturerAccess()</i> = 0xF082: 2
AAaa	2's comp	AFE CELL Map
BBbb	2's comp	Cell Voltage 1
CCcc	2's comp	Cell Voltage 2
DDdd	2's comp	Cell Voltage 3
EEee	2's comp	Cell Voltage 4
FFff	2's comp	Cell Voltage 5
GGgg	2's comp	Cell Voltage 6
HHhh	2's comp	Cell Voltage 7
IIii	2's comp	Cell Voltage 8
JJjj	2's comp	Cell Voltage 9
KKkk	2's comp	Cell Voltage 10
LLll	2's comp	Cell Voltage 11
MMmm	2's comp	Cell Voltage 12
NNnn	2's comp	Cell Voltage 13
OOoo	2's comp	Cell Voltage 14

For 0xF082, the *ManufacturerData()* output format is:
ZZYYaaAAabbBBccCCddDDeeEEffFFggGGhhHHiIlJjJkkKK, where:

Value	Format	Description
ZZ	Byte	8-bit counter, increments when raw ADC values are refreshed (every 250 ms)
YY	Byte	Output Status <i>ManufacturerAccess()</i> = 0xF081: 1 <i>ManufacturerAccess()</i> = 0xF082: 2
AAaa	2's comp	AFE CELL Map
BBbb	2's comp	Cell Voltage 15
CCcc	—	Reserved
DDdd	2's comp	Ext Ave Cell Voltage
EEeeFFff	2's comp	VAUX Voltage
GGgg	2's comp	Current Coulomb Counter
HHhh	2's comp	TS1 Temperature
IIii	2's comp	TS2 Temperature
JJjj	2's comp	TS3 Temperature
KKkk	2's comp	Internal Gauge Temperature

12.2.1 Cell Voltage Calibration

- Apply known voltages in mV to the cell voltage inputs. See the companion AFE data manual, *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications (SLUSBK2)*, for the actual physical pin connections:
 - V_{CELL1} is the lowest physical cell in the stack.
 - V_{CELL2} is the next (second) cell in the stack.
 - V_{CELLn} is the top cell in the stack.
- If *ManufacturerStatus()[CAL]* = 0, send 0x002D to *ManufacturerAccess()* to enable the [CAL] flag.
- Send 0xF081 or 0xF082 to *ManufacturerAccess()* to enable cell voltage output on *ManufacturerData()*, depending on which cells are being calibrated.
- Poll *ManufacturerData()* until the 8-bit counter value increments by 2 before reading data.
- See the readings of factory calibrated cell voltages from *ManufacturerData()* beginning with Cell 1 and ending in the number of configured cell, n, with the maximum being 15. Depending on the number of cells, the data is available through two separate read-blocks (0xF081 or 0xF082):
 - $FCAL_{CELL1} = BBbb$ of *ManufacturerData()*
 - $FCAL_{CELL10} = KKkk$ of *ManufacturerData()*.
 - $FCAL_{CELL15} = BBbb$ of *ManufacturerData()*.
- Average several readings for higher accuracy. Poll *ManufacturerData()* until ZZ increments, to indicate that updated values are available:
 - $FCAL_{CELLx} = [FCAL_{CELLx}(\text{reading } n) + \dots + FCAL_{CELLx}(\text{reading } 1)]/n$
- Calculate Cell n Offset value: where N = number of cells.
 - $FCAL_{CELLn} - \text{Reference Cell Voltage} = \text{Cell } n \text{ Offset}$
- Write the new **Cell n Offset** value to data flash.
- Re-check the voltage reading and if it is not accurate, repeat Steps 5 through 8.
- Send 0x002D to *ManufacturerAccess()* to clear the [CAL] flag if all calibration is complete.

12.2.2 External Average Voltage Calibration

The BQ78350-R1A can be configured with an external resistor divider to measure the battery stack voltage directly. This measurement has its own calibration procedure.

1. Apply a known voltage in mV to the battery terminals on which the resistor divider is connected.
2. Divide the known battery voltage by the number of cells configured in **AFE Cell Map**. This would be Actual Avg Cell Voltage.
3. Read *ExtAveCellVoltage()*.
4. Calculate the Ext Cell Divider Gain.
 - New **Ext Cell Divider Gain** = (Old **Ext Cell Divider Gain** × Actual Avg Cell Voltage)/Measured Avg Cell Voltage
5. Update **Ext Cell Divider Gain** in data flash.

12.2.3 VAUX Voltage Calibration

The BQ78350-R1A can be configured with an auxiliary voltage measurement input. This measurement has its own calibration procedure, as follows:

1. Apply a known voltage in mV to the VAUX input.
2. Read *VAUXVoltage()*.
3. Calculate the VAUX Gain.
 - New **VAUX Gain** = (Old **VAUX Gain**/VAUX Voltage)
4. Update **VAUX Gain** in data flash.

12.2.4 Voltage Calibration Data Flash

Class	Subclass	Name	Type	Length in Bytes	Min	Max	Default	Unit
Calibration	Voltage	Cell1 Offset to Cell15 Offset	Integer	1	-128	127	0 ⁽¹⁾	—
Calibration	VAUX Voltage	VAUX Gain	Integer	4	0	10000	5000	—
Calibration	Ext Cell Voltage	Ext Ave Divider Gain	Integer	2	0	32767	5000	—

⁽¹⁾ Setting this value to 0 causes the gauge to use the internal factory calibration default.

12.2.5 Current Calibration

12.2.5.1 CC Offset Calibration

1. Apply a known current of 0 mA, and ensure no current is flowing through the sense resistor connected between the SRP and SRN pins.
2. If *ManufacturerStatus()[CAL]* = 0, send 0x002D to *ManufacturerAccess()* to enable the [CAL] flag.
3. Send 0xF082 to *ManufacturerAccess()* to enable factory calibrated CC output on *ManufacturerData()*.
4. Poll *ManufacturerData()* until ZZ increments by 2 before reading data.
5. Obtain the factory calibrated conversion readings of current from *ManufacturerData()*:
 - $FCAL_{CC} = \text{GGgg of } ManufacturerData()$
 - Is $FCAL_{CC} < 0x8000$? If yes, use $FCAL_{CC}$; otherwise, $FCAL_{CC} = -(0xFFFF - AAaa + 0x0001)$.
6. Average several readings for higher accuracy. Poll *ManufacturerData()* until ZZ increments to indicate that updated values are available:
 - $FCAL_{CC} = [FCAL_{CC}(\text{reading } n) + \dots + FCAL_{CC}(\text{reading } 1)]/n$
7. Read *Coulomb Counter Offset Samples* from data flash.
8. Calculate offset value:
 - CC offset = $FCAL_{CC} \times (\text{Coulomb Counter Offset Samples})$
9. Write the new *CC Offset* value to data flash.
10. Re-check the current reading and if it is not accurate, repeat the steps.
11. Send 0x002D to *ManufacturerAccess()* to clear the [CAL] flag if all calibration is complete.

12.2.5.2 CC Gain/Capacity Gain Calibration

1. Apply a known current, which is the nominal discharge current of the battery and can be up to 100 A.
2. If *ManufacturerStatus()*[CAL] = 0, send 0x002D to *ManufacturerAccess()* to enable the [CAL] flag.
3. Send 0xF082 to *ManufacturerAccess()* to enable factory calibrated CC output on *ManufacturerData()*.
4. Poll *ManufacturerData()* until ZZ increments by 2 before reading data.
5. Using the *ManufacturerAccess()* Commands 0x001F and 0x0020, turn ON the CHG and DSG FETs.
6. Read the factory calibrated conversion readings of current from *ManufacturerData()*:
 - $FCAL_{CC} = GG_{gg}$ of *ManufacturerData()*
 - Is $FCAL_{CC} < 0x8000$? If yes, use $FCAL_{CC}$; otherwise, $FCAL_{CC} = -(0xFFFF - GG_{gg} + 0x0001)$.
7. Average several readings for higher accuracy. Poll *ManufacturerData()* until ZZ increments to indicate that updated values are available:
 - $FCAL_{CC} = [FCAL_{CC}(\text{reading } n) + \dots + FCAL_{CC}(\text{reading } 1)]/n$
8. Using the *ManufacturerAccess()* Commands 0x001F and 0x0020, turn OFF the CHG and DSG. FETs
9. Read **Coulomb Counter Offset Samples** from data flash.
10. Calculate gain values:

$$CC\ Gain = \frac{I_{CC}}{ADC_{CC} - \frac{CC\ Offset}{Coulomb\ Counter\ Offset\ Samples}}$$

$$Capacity\ Gain = CC\ Gain \times 298261.6178$$

11. Write the new **CC Gain** and **Capacity Gain** values to data flash.
12. Re-check the current reading. If the reading is not accurate, repeat the steps.
13. Send 0x002D to *ManufacturerAccess()* to clear the [CAL] flag if all calibration is complete.

12.2.6 Deadbands

The BQ78350-R1A can be configured to ignore current and coulomb measurements below individually programmable levels.

12.2.6.1 Current Deadband

When *Current()* measures a value less than the value programmed in **Current Deadband**, *Current()* will report 0. This has no effect on the coulomb counting for the gas gauging functionality.

The value of **Current Deadband** should be selected based on the characterization of the battery electronics design combined with the environment in which the battery will be used. If the PCB senses noise causing a real no-current condition to report a non-zero value, then **Current Deadband** could be adjusted accordingly.

12.2.6.2 Coulomb Counter Deadband

During normal operation, there could be noise generated in the battery electronics environment that could cause the BQ78350-R1A to accumulate incorrectly (positively or negatively). To filter out this noise, the **Coulomb Counter Deadband** setting is used. Any input below this threshold is not accumulated.

12.2.7 Current Calibration Data Flash

Class	Subclass	Name	Format	Length in Bytes	Min	Max	Default	Unit
Calibration	Current	CC Gain	Float	4	1.00E-001	9.00E+000	8.4381	mΩ
Calibration	Current	Capacity Gain	Float	4	2.98E+004	2.69E+07	2516761.36	—
Calibration	Current Offset	CC Offset	Integer	2	-32768	32767	0	—
Calibration	Current Offset	Coulomb Counter Offset Samples	Unsigned Integer	2	0	65535	64	—

Class	Subclass	Name	Format	Length in Bytes	Min	Max	Default	Unit
Calibration	Current Deadband	Deadband	Unsigned Integer	1	0	255	3	mA
Calibration	Current Deadband	Coulomb Counter Deadband	Unsigned Integer	1	0	255	38	264 nV

12.2.8 Temperature Calibration

12.2.8.1 TS1–TS2–TS3 Calibration

1. Apply a known temperature in 0.1°C, and ensure that temperature $TEMP_{TSx}$ is applied to the thermistor connected to the TSx pin. "TSx" refers to TS1, TS2, or TS3, whichever is applicable.
2. Send 0xF082 to *ManufacturerAccess()* to enable factory calibrated Temperature output on *ManufacturerData()*.
3. Poll *ManufacturerData()* until ZZ increments by 2 before reading data.
4. Read the factory calibrated conversion readings of Temperature from *ManufacturerData()*:
 - $FCAL_{TS1} = \text{HHhh of } ManufacturerData()$
 - $FCAL_{TS2} = \text{Illi of } ManufacturerData()$
 - $FCAL_{TS3} = \text{JJjj of } ManufacturerData()$
5. Read the TSx offset_{old} from **External × Temp Offset**, where × is 1, 2, or 3.
6. Re-check the temperature reading. If the reading is not accurate, repeat the steps.
7. Calculate the temperature offset:

$$TSx \text{ offset} = TEMP_{TSx} - TSx + TSx \text{ offset}_{old}$$
 where × is 1, 2, or 3.
8. Write the new **External × Temp Offset** (where × is 1, 2, or 3) value to data flash.
9. Re-check the temperature reading. If the reading is not accurate, repeat the steps.

12.2.9 Temperature Calibration Data Flash

Class	Subclass	Name	Type	Length in Bytes	Min	Max	Default	Unit
Calibration	Temperature	External 1 Temp Offset	Integer	1	–128	127	0	0.1 °C
Calibration	Temperature	External 2 Temp Offset	Integer	1	–128	127	0	0.1 °C
Calibration	Temperature	External 3 Temp Offset	Integer	1	–128	127	0	0.1 °C
Calibration	Temperature	Gauge Internal Temp Offset	Integer	1	–128	127	0	0.1 °C

12.2.10 External Temp Model

Class	Subclass	Name	Type	Length in Bytes	Min	Max	Default	Unit
Calibration	Cell Temp Model	Coefficient a1	Integer	2	–32768	32768	–11130	—
Calibration	Cell Temp Model	Coefficient a2	Integer	2	–32768	32768	19142	—
Calibration	Cell Temp Model	Coefficient a3	Integer	2	–32768	32768	–19262	—
Calibration	Cell Temp Model	Coefficient a4	Integer	2	–32768	32768	28203	—
Calibration	Cell Temp Model	Coefficient a5	Integer	2	–32768	32768	892	—
Calibration	Cell Temp Model	Coefficient b1	Integer	2	–32768	32768	328	—
Calibration	Cell Temp Model	Coefficient b2	Integer	2	–32768	32768	–605	—
Calibration	Cell Temp Model	Coefficient b3	Integer	2	–32768	32768	–2443	—
Calibration	Cell Temp Model	Coefficient b4	Integer	2	–32768	32768	4969	—
Calibration	Cell Temp Model	Rc0	Integer	2	–32768	32768	11703	Ω
Calibration	Cell Temp Model	Adc0	Integer	2	–32768	32768	11703	—

Display Port

13.1 Introduction

The Display Port feature can provide a visual display of *RelativeStateOfCharge()* or *AbsoluteStateOfCharge()*, and can be activated in several ways. This feature can use LEDs or a 5-bar LCD and has a variety of configuration options to enable a wide range of indications.

The BQ78350-R1A display type is set in **LED Configuration [LCDEN]**. When [LCDEN] = 0, the display type is LED; when [LCDEN] = 1, the display type is LCD.

13.1.1 Light Emitting Diode (LED) Display Operation

The LED display is the default display type for the BQ78350-R1A. When the LED display is activated, the device turns on the appropriate LEDs through the LED1..5 pins when a push button is pressed or a command is sent to the device.

13.1.2 Liquid Crystal Display (LCD) Operation

The LCD controller supports 3- to 5-segment static bar graph liquid crystal displays (LCDs). The LCD is operational at all times except when the BQ78350-R1A is in the SHUTDOWN power mode.

A static LCD generally has one large electrode on one side of the liquid crystal material called a "common," and a number of smaller electrodes on the other side called "segments." Segments are made visible (black) by applying a differential voltage between the back plane signal of the LCD and the corresponding segment pin. Segments are turned off when there is no voltage difference between the back plane signal and a segment signal. The display signals must be periodically reversed to ensure zero average DC voltage and to refresh the display.

Liquid crystal displays having an operating voltage range of 2.5 V to 6 V and a refresh frequency between 30 Hz and 200 Hz are supported. The display refresh must be implemented such that the device current consumption requirements during sleep and active modes are not violated. The display refresh frequency must always be set as low as the LCD specification allows in order to minimize current consumption.

Static LCD drive procedure is as follows:

- Step 1. Drive back plane to ground, drive "on" segments high, "off" segments to ground.
- Step 2. Wait for time 1/refresh frequency.
- Step 3. Drive back plane high, drive "on" segments to ground, "off" segments high.
- Step 4. Wait for time 1/refresh frequency.
- Step 5. Go to Step 1.

13.2 Display Activation

13.2.1 LED Display Activation

The LED display is activated for a period of **LED Hold Time**:

- If the $\overline{\text{DISP}}$ pin is low for < **LED Hold Time**.
- If [LEDR] = 1 AND the device is Reset.
- If [LEDCHG] = 1 AND *Current()* > 0.
- If *ManufacturerStatus()* = 0x002C.

If *SafetyStatus()* [CUV] flag is set, the display is disabled.

If **[LEDRCA]** = 1 and the *BatteryStatus()* **[RCA]** change from 0 to 1, the LED display will also flash at the **LED Flash Rate** according to the **LED Flash Alarm**.

13.2.2 LCD Display Activation

The LCD is activated all the time during operation except:

- If *SafetyStatus()* **[CUV]** flag is set.
- If the device is in the SHUTDOWN power mode.
- If LCD_SLEEP_DIS is set AND the device is in the SLEEP power mode.

13.3 State-Of-Charge Display

The state-of-charge can display *RelativeStateOfCharge()* or *(AbsoluteStateOfCharge()/DesignCapacity())*, selectable with **LED Configuration [LEDMODE]**.

LED/LCD Segment Output ON	State-Of-Charge
LED1/LCD1	LED Thresh 1
LED2/LCD2	LED Thresh 2
LED3/LCD3	LED Thresh 3
LED4/LCD4	LED Thresh 4
LED5/LCD5	LED Thresh 5

Table 13-1. Display Thresholds

Class	Subclass	Name	Format	Length in Bytes	Min	Max	Default
LED Support	LED Config	LED Flash Alarm	Integer	1	0%	100%	10%
LED Support	LED Config	LED Thresh 1	Integer	1	0%	100%	0%
LED Support	LED Config	LED Thresh 2	Integer	1	0%	100%	20%
LED Support	LED Config	LED Thresh 3	Integer	1	0%	100%	40%
LED Support	LED Config	LED Thresh 4	Integer	1	0%	100%	60%
LED Support	LED Config	LED Thresh 5	Integer	1	0%	100%	80%

The default settings are for a 5-LED/LCD segment display; however, if fewer LEDs/LCD segments are required, that is, when extra Host Controlled GPIOs are required, then less LEDs/LCD segments can be used. In this case, the lower LEDs/LCD segments should be used with the unused LEDs/LCD segments being set to 100%. For example, in a 3-LED case, LED1, LED2, and LED3 should be used and can be configured for 0, 33%, and 66%, respectively, with LED4 and LED5 set to 100%.

NOTE: Unused LED settings must be set to 100% even if the pin is not used.

13.4 LED and LCD Display Configuration

All data flash settings are available through the **LED Support:LED Config** subclass. When the display is enabled for LEDs, the following configuration options are available:

- **LED Blink Period**—During charging, the top LED segment flashes with the LED Blink Period time period, and lower LEDs will be fully ON. For example, if battery charge is 36% and the display uses five LEDs, LED 1 will be ON and LED 2 will blink. **[LEDRCA]** will override this setting if active.
- **LED Flash Period**—During discharge alarm, the remaining LED segments flash with the **LED Flash Period** time period: for example, if battery charge is 36% and the display uses five LEDs, LED 1 and LED 2 will blink.
- **LED Delay**—An activation delay from one LED to another LED can be set with this value. For LCD, this configuration is not used because the display is always active.
- **LED Hold Time**—After display activation, the display will stay on for the **LED Hold Time** period. For

LCD, this configuration is not used because the display is always active.

Table 13-2. LED Configuration Values

Class	Subclass	Name	Format	Length in Bytes	Min	Max	Default	Unit
LED Support	LED Config	LED Flash Period	Unsigned Integer	2	32	65535	512	488 μ s
LED Support	LED Config	LED Blink Period	Unsigned Integer	2	32	65535	1024	488 μ s
LED Support	LED Config	LED Delay	Unsigned Integer	2	32	65535	100	488 μ s
LED Support	LED Config	LED Hold Time	Unsigned IntegerU1	1	0	63	16	0.25 s

13.5 LCD Specific Display Configuration

When the display is enabled for a 5-bar LCD, then the following configuration options are available.

The **LCD Refresh** parameter is the LCD refresh frequency setting register. If the LCD display "blinks," or is not constantly on, then this value should be reduced.

Table 13-3. LCD Configuration Values

Class	Subclass	Name	Format	Length in Bytes	Min	Max	Default	Unit
LED Support	LED Config	LCD Refresh	Unsigned Integer	1	20	100	35	Hz

13.6 LED Configuration Register

This register contains a variety of display enable/display settings.

Table 13-4. Display Configuration Enable/Disable Options

Class	Subclass	Name	Format	Min	Max	Default	Unit
Settings	Configuration	LED Configuration	Hex	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
RSVD	RSVD	LCD_SLEEP_DIS	LCDEN	LEDMODE	LEDCHG	LEDRCA	LEDR

RSVD (Bits 7–6): Reserved. Do not use.

LCD_SLEEP_DIS (Bit 5): Disable LCD Display when in SLEEP

1 = LCD display is disabled during SLEEP.

0 = LED display is enabled during SLEEP (default).

LCDEN (Bit 4): LCD Display Type Enable

1 = LCD display type is selected.

0 = LED display type is selected (default).

LEDMODE (Bit 3): Determines if *RelativeStateOfCharge()* or *AbsoluteStateOfCharge()* is used for the display.

1 = *AbsoluteStateOfCharge()* is used.

0 = *RelativeStateOfCharge()* is used (default).

LEDCHG (Bit 2): Enables LED during charging. For LCD, the display is always active so this setting has no effect.

- 1 = Activates display due to charging current
- 0 = Does not activate display due to charging current (default)

LEDRCA (Bit 1): Enables flashing of the LED display when the *[RCA]* flag in *BatteryStatus()* *BatteryStatus* is set.

- 1 = If the LED display is activated when *[RCA]* is set, the display flashes with **LED Flash Period**.
- 0 = The LED display is not activated due to *[RCA]* being set. (default)

LEDR (Bit 0): Enables activation of the LED display on device-reset exit. For LCD, the display is activated on device-reset exit; same as setting **LEDR** = 1.

- 1 = LED display is not activated on exit from device reset.
- 0 = LED display is activated (simulates a $\overline{\text{DISP}}$ transition) on exit from device reset (default).

Host Controlled GPIO

14.1 Introduction

The BQ78350-R1A can have the SMBus host read or drive GPIO. Two of the available seven GPIO are dedicated GPIO (GPIO A and GPIO B), and the other five are default configured as the LED display (LED1...5). However, each LED pin can be individually selected to be read or driven by the host SMBus as a GPIO.

14.2 Configuring the GPIO

Each pin chosen as a host controlled GPIO pin must be selected as a GPIO, even the dedicated ones, in **GPIO Config**. Once selected, the Input or Output selection is set in **GPIO Output Enable**. If the corresponding bit in **GPIO Config** is not set then the bit in **GPIO Output Enable** is ignored. If configured as an output the default state upon reset of the device can be set through **GPIO Output Default**. Additionally each pin can be configured as either Open Drain (OD) or as a 3-mA Current Sink (CS) through the **GPIO Type** settings.

Table 14-1. Host Controlled GPIO Enable Options

Class	Subclass	Name	Format	Min	Max	Default	Unit
GPIO	GPIO Config	GPIO Config	Hex	0x00	0xFE	0x00	—

7	6	5	4	3	2	1	0
GPIO B	GPIO A	LED5	LED4	LED3	LED2	LED1	RSVD

GPIO B (Bit 7): GPIO B

- 1 = Active as GPIO
- 0 = Inactive

GPIO A (Bit 6): GPIO A

- 1 = Active as GPIO
- 0 = Inactive

LED5 (Bit 5): GPIO 5

- 1 = Active as GPIO
- 0 = Inactive (used as LED5)

LED4 (Bit 4): GPIO 4

- 1 = Active as GPIO
- 0 = Inactive (used as LED4)

LED3 (Bit 3): GPIO 3

- 1 = Active as GPIO
- 0 = Inactive (used as LED3)

LED2 (Bit 2): GPIO 2

- 1 = Active as GPIO
- 0 = Inactive (used as LED2)

LED1 (Bit 1): GPIO 1

1 = Active as GPIO

0 = Inactive (used as LED1)

RSVD (Bit 0): Reserved
Table 14-2. Host Controlled GPIO Output Enable Configuration

Class	Subclass	Name	Format	Min	Max	Default	Unit
GPIO	GPIO Config	GPIO Output Enable	Hex	0x00	0xFE	0x00	—

7	6	5	4	3	2	1	0
GPIO B	GPIO A	LED5	LED4	LED3	LED2	LED1	RSVD

GPIO B (Bit 7): GPIO B

1 = Output

0 = Input

GPIO A (Bit 6): GPIO A

1 = Output

0 = Input

LED5 (Bit 5): GPIO 5

1 = Output

0 = Input

LED4 (Bit 4): GPIO 4

1 = Output

0 = Input

LED3 (Bit 3): GPIO 3

1 = Output

0 = Input

LED2 (Bit 2): GPIO 2

1 = Output

0 = Input

LED1 (Bit 1): GPIO 1

1 = Output

0 = Input

RSVD (Bit 0): Reserved
Table 14-3. Host Controlled GPIO Output Default Configuration

Class	Subclass	Name	Format	Min	Max	Default	Unit
GPIO	GPIO Config	GPIO Output Default	Hex	0x00	0xFE	0x00	—

7	6	5	4	3	2	1	0
GPIO B	GPIO A	LED5	LED4	LED3	LED2	LED1	RSVD

GPIO B (Bit 7): GPIO B

1 = If enabled as output, high

0 = If enabled as output, low

GPIO A (Bit 6): GPIO A

1 = If enabled as output, high

0 = If enabled as output, low

LED5 (Bit 5): GPIO 5

1 = If enabled as output, high

0 = If enabled as output, low

LED4 (Bit 4): GPIO 4

1 = If enabled as output, high

0 = If enabled as output, low

LED3 (Bit 3): GPIO 3

1 = If enabled as output, high

0 = If enabled as output, low

LED2 (Bit 2): GPIO 2

1 = If enabled as output, high

0 = If enabled as output, low

LED1 (Bit 1): GPIO 1

1 = If enabled as output, high

0 = If enabled as output, low

RSVD (Bit 0): Reserved

Table 14-4. Host Controlled GPIO Type Configuration

Class	Subclass	Name	Format	Min	Max	Default	Unit
GPIO	GPIO Config	GPIO Type	Hex	0x00	0xFE	0xC0	—

7	6	5	4	3	2	1	0
GPIO B	GPIO A	LED5	LED4	LED3	LED2	LED1	RSVD

GPIO B (Bit 7): GPIO B

1 = Open Drain

0 = 3-mA Current Sink

GPIO A (Bit 6): GPIO A

1 = Open Drain

0 = 3-mA Current Sink

LED5 (Bit 5): GPIO 5

1 = Open Drain

0 = 3-mA Current Sink

LED4 (Bit 4): GPIO 4

1 = Open Drain

0 = 3-mA Current Sink

LED3 (Bit 3): GPIO 3

- 1 = Open Drain
- 0 = 3-mA Current Sink

LED2 (Bit 2): GPIO 2

- 1 = Open Drain
- 0 = 3-mA Current Sink

LED1 (Bit 1): GPIO 1

- 1 = Open Drain
- 0 = 3-mA Current Sink

RSVD (Bit 0): Reserved

14.3 Using the GPIO

The status of all enabled GPIO can be read through *GPIOStatus()*, and the enabled outputs can be driven to a specific state through *GPIOControl()*. When enabling a mix of the LED and GPIO pins to be used as host controlled GPIO, care should be taken to ensure they are configured correctly for appropriate desired operation.

Key Input

15.1 Introduction

The BQ78350-R1A uses the KEYIN input to enable or disable the DSG FET if safety conditions allow.

15.2 Input Configuration

The polarity of KEYIN detection can be set to active high or active low. If the KEYIN driver does not drive to both high and low states then the KEYIN pin will require an external pullup, typ 100 k, to VCC. If **FET Options** [KEY_POL] = 0, then the KEYIN input is active low; if [KEY_POL] = 1, then the KEYIN input is active high. To enable this feature, **FET Options** [KEYEN] must be set.

15.3 Operation

When [KEY_POL] = 0 (active low) and the KEYIN input is low, then the BQ78350-R1A operates normally. However, if KEYIN were to transition to a high state and remain in that state for KEYIN Time, then the BQ78350-R1A would control the companion AFE to turn off the DSG FET. If the KEYIN input transitions back to low before **KEYIN Time** expires, then the BQ78350-R1A continues to operate normally and the BQ78350-R1A is not influenced by the KEYIN pin.

If the KEYIN input transitions back to a low state and remains in that state for **KEYIN Time**, then the BQ78350-R1A would control the companion AFE to turn on the DSG FET only if all other safety conditions allow.

If **KEYEN** = 1 and if the BQ78350-R1A experiences a full power-on reset, then DSG FET will be turned OFF and the KEYIN transition is again required to turn on the DSG FET.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Power	Keyin	Time	Integer	1	0	10	2	s

Communications

16.1 Introduction

The BQ78350-R1A uses SMBus v1.1 with MASTER mode and packet error checking (PEC) options per the SBS specification.

16.2 SMBus On and Off State

The BQ78350-R1A detects an SMBus off state when SMBC and SMBD are logic-low for ≥ 2 seconds. Clearing this state requires either SMBC or SMBD to transition high. Within 1 ms, the communication bus is available.

16.3 Packet Error Checking

The BQ78350-R1A can receive or transmit data with or without packet error checking (PEC).

In the write-word protocol, if the host does not support PEC, the last byte of data is followed by a stop condition and the **[HPE]** bit should be set to 0 (default).

In the write-word protocol, the BQ78350-R1A receives the PEC after the last byte of data from the host. If the host does not support PEC, the last byte of data is followed by a stop condition. After receipt of the PEC, the BQ78350-R1A compares the value to its calculation. If the PEC is correct, the BQ78350-R1A responds with an ACKNOWLEDGE. If it is not correct, the BQ78350-R1A responds with a NOT ACKNOWLEDGE and sets an error code. If the host supports PEC, the **[HPE]** bit should be set to 1.

In the read-word and block-read in MASTER mode, the host generates an ACKNOWLEDGE after the last byte of data sent by the BQ78350-R1A. The BQ78350-R1A then sends the PEC, and the host, acting as a master-receiver, generates a NOT ACKNOWLEDGE and a stop condition.

16.4 Slave Address

The BQ78350-R1A has a configurable addressing scheme that can be enabled or this feature can be disabled resulting in the slave address being fixed as 0x16/0x17.

When **[FIXED_ADDR]** in **SMB Configuration** is clear (0), then the slave address is determined by the voltage measured at the SMBA pin. The voltage on the SMBA pin is created via either being tied to VCC, VSS, or through an external resistor divider. The external divider can be enabled and disabled via the ADREN (pin 29) and an external FET. The upper resistor should be connected between VCC and SMBA with the lower resistor of the divider connected between SMBA and VSS. Both of these resistors are recommended to be 1% tolerance or better.

Upon exit from Power On Reset (POR) or when *OperationStatus()* **[PRES]** transitions from 0 to 1, the BQ78350-R1A drives ADREN high, takes a number of sequential voltage measurements (set by **Addr Reads**) of the SMBA pin taking approximately 32 ms each. The corresponding address, set by **SMBTAR_ADDR0...7**, is determined for each measurement with the most common address selection being the one used. If all are different, then the average voltage value is used to determine the address. Upon completion of the address selection, ADREN is set low to turn off the resistor divider to conserve power.

Care should be taken in the setting of **Addr Reads** as the BQ78350-R1A will only respond to address 0x16/0x17 until at least **Addr Read** $\times$ 32 ms after POR.

The actual address corresponding to the SMBA voltage is configurable per the following table.

SMBA Pin Voltage			
Channel	VSMBAMIN Voltage (V)	VSMBAMAX Voltage (V)	Address Data Flash
0	0.070	0.130	SMBTAR_ADDR0
1	0.170	0.230	SMBTAR_ADDR1
2	0.270	0.330	SMBTAR_ADDR2
3	0.370	0.430	SMBTAR_ADDR3
4	0.470	0.530	SMBTAR_ADDR4
5	0.570	0.630	SMBTAR_ADDR5
6	0.670	0.730	SMBTAR_ADDR6
7	0.770	0.830	SMBTAR_ADDR7
0	All other values		0x16

If **SMB Configuration [FIXED_ADDR]** is set (1), then the slave address is 0x16/0x17 and the ADREN and SMBA pins are ignored and not used.

NOTE: When determining the values for SMBTAR_ADDR0...7, take into account the R/W bit by programming all addresses to even numbers.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	Addr Reads	Integer	1	0	10	3	—
Settings	Configuration	SMBTAR_ADDR0	Integer	1	0x00	0xFE	0x20	—
Settings	Configuration	SMBTAR_ADDR1	Integer	1	0x00	0xFE	0x22	—
Settings	Configuration	SMBTAR_ADDR2	Integer	1	0x00	0xFE	0x24	—
Settings	Configuration	SMBTAR_ADDR3	Integer	1	0x00	0xFE	0x25	—
Settings	Configuration	SMBTAR_ADDR4	Integer	1	0x00	0xFE	0x26	—
Settings	Configuration	SMBTAR_ADDR5	Integer	1	0x00	0xFE	0x28	—
Settings	Configuration	SMBTAR_ADDR6	Integer	1	0x00	0xFE	0x2A	—
Settings	Configuration	SMBTAR_ADDR7	Integer	1	0x00	0xFE	0x2C	—

16.5 Broadcasts to Smart Charger and Smart Battery Host

If the **[HPE]** bit is enabled, MASTER mode broadcasts to the host address are PEC enabled. If the **[CPE]** bit is enabled, MASTER mode broadcasts to the smart-charger address are PEC enabled. The **[BCAST]** bit enables all broadcasts to a host or a smart charger. When the **[BCAST]** bit is enabled, the following broadcasts are sent:

- *ChargingVoltage()* and *ChargingCurrent()* broadcasts are sent to the smart-charger device address (**Charger Address**) periodically. The default period is set in **Charger Request Timer**.
- If any of the **[OCA]**, **[TCA]**, **[OTA]**, **[TDA]**, **[RCA]**, **[RTA]** flags are set, the *AlarmWarning()* broadcast is sent to the host device address (**Host Address**) at the period set in **Alarm Timer**. Broadcasts stop when all flags above have been cleared.
- If any of the **[OCA]**, **[TCA]**, **[OTA]**, **[TDA]** flags are set, the *AlarmWarning()* broadcast is sent to a smart-charger device address (**Charger Address**) at the period set in **Alarm Timer**. Broadcasts stop when all flags above have been cleared.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	SMB Master Mode	Host Address	Integer	1	0x00	0xFE	10	—
Settings	SMB Master Mode	Charger Address	Integer	1	0x00	0xFE	12	—
Settings	SMB Master Mode	Alarm Timer	Integer	1	0	255	10	s

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	SMB Master Mode	Charger Request Timer	Integer	1	0	255	50	s

16.6 SMB Configuration Options

This register configures various SMBus related features.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	Smb Configuration	hex	1	0x00	0xB7	0xA0	—

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Low Byte	FIXED_ADDR	RSVD	BLT1	BLT0	RSVD	HPE	CPE	BCAST

FIXED_ADDR (Bit 7): This bit enables the BQ78350-R1A to determine its slave address via measurement of the SMBA pin or to use a fixed address of 0x16.

0 = The SMBus slave address is selected via SMBA.

1 = The SMBus address is 0x16. (default)

RSVD (Bit 6): Reserved

BLT1:0 (Bits 5–4): Bus low timeout

0,0 = No SBS bus low timeout

0,1 = 1-s SBS bus low timeout

1,0 = 2-s SBS bus low timeout (default)

1,1 = 3-s SBS bus low timeout

RSVD (Bit 3): Reserved

HPE (Bit 2): PEC on host communication

0 = Disabled (default)

1 = Enabled

CPE (Bit 1): PEC on charger communication

0 = Disabled (default)

1 = Enabled

BCAST (Bit 0): Enable alert and charging broadcast from device to host

0 = Disabled (default)

1 = Enabled

SBS Commands

17.1 Summary

Table 17-1. SBS Commands Summary

SBS Cmd	Mode	Name	Format	Size in Bytes	Min	Max	Default	Unit
0x00	R/W	ManufacturerAccess	hex	2	0x0000	0xFFFF	—	
0x01	R/W	RemainingCapacityAlarm	unsigned int	2	0	65535	300	mAh or 10 mWh
0x02	R/W	RemainingTimeAlarm	unsigned int	2	0	65535	10	min
0x03	R/W	BatteryMode	hex	2	0x0000	0xFFFF	—	
0x04	R/W	AtRate	integer	2	–32768	32767	—	mAh or 10 mWh
0x05	R	AtRateTimeToFull	unsigned int	2	0	65534	—	min
0x06	R	AtRateTimeToEmpty	unsigned int	2	0	65534	—	min
0x07	R	AtRateOK	unsigned int	2	0	65535	—	
0x08	R	Temperature	unsigned int	2	0	65535	—	0.1 K
0x09	R	Voltage	unsigned int	2	0	65535	—	mV
0x0A	R	Current	integer	2	–32768	32767	—	mA
0x0B	R	AverageCurrent	integer	2	–32768	32767	—	mA
0x0C	R	MaxError	unsigned int	1	0%	100%	—	
0x0D	R	RelativeStateOfCharge	unsigned int	1	0%	100%	—	
0x0E	R	AbsoluteStateOfCharge	unsigned int	1	0%	100+%	—	
0x0F	R/W	RemainingCapacity	unsigned int	2	0	65535	—	mAh or 10 mWh
0x10	R	FullChargeCapacity	unsigned int	2	0	65535	—	mAh or 10 mWh
0x11	R	RunTimeToEmpty	unsigned int	2	0	65534	—	min
0x12	R	AverageTimeToEmpty	unsigned int	2	0	65534	—	min
0x13	R	AverageTimeToFull	unsigned int	2	0	65534	—	min
0x14	R	ChargingCurrent	unsigned int	2	0	65534	—	mA
0x15	R	ChargingVoltage	unsigned int	2	0	65534	—	mV
0x16	R	BatteryStatus	unsigned int	2	0x0000	0xdbff	—	
0x17	R	CycleCount	unsigned int	2	0	65535	—	
0x18	R	DesignCapacity	unsigned int	2	0	65535	—	mAh or 10 mWh
0x19	R	DesignVoltage	unsigned int	2	0	65535	—	mV
0x1A	R	SpecificationInfo	hex	2	0x0000	0xFFFF	0x0031	
0x1B	R	ManufacturerDate	unsigned int	2	0	65535	—	
0x1C	R	SerialNumber	hex	2	0x0000	0xFFFF	0x0001	
0x20	R	ManufacturerName	string	11+1	—	—	Texas Instruments	ASCII
0x21	R	DeviceName	string	7+1	—	—	BQ78350-tbd	ASCII
0x22	R	DeviceChemistry	string	4+1	—	—	LiON	ASCII
0x23	R	ManufacturerData	String	14+1	—	—	—	ASCII
0x2B	R/W	HostFETControl	hex	2	0	65535	—	
0x2C	R	GPIOStatus	hex	2	0	65535	—	

Table 17-1. SBS Commands Summary (continued)

SBS Cmd	Mode	Name	Format	Size in Bytes	Min	Max	Default	Unit
0x2D	R/W	GPIOControl	hex	2	0	65535	—	
0x2E	R	VAUXVoltage	unsigned int	2	0	65535	1	
0x2F	R/W	Authenticate	hex	20+1	—	—	—	
0x30	R	Reserved	—	—	—	—	—	—
0x31	R	CellVoltage15	unsigned int	2	0	65535	—	mV
0x32	R	CellVoltage14	unsigned int	2	0	65535	—	mV
0x33	R	CellVoltage13	unsigned int	2	0	65535	—	mV
0x34	R	CellVoltage12	unsigned int	2	0	65535	—	mV
0x35	R	CellVoltage11	unsigned int	2	0	65535	—	mV
0x36	R	CellVoltage10	unsigned int	2	0	65535	—	mV
0x37	R	CellVoltage9	unsigned int	2	0	65535	—	mV
0x38	R	CellVoltage8	unsigned int	2	0	65535	—	mV
0x39	R	CellVoltage7	unsigned int	2	0	65535	—	mV
0x3A	R	CellVoltage6	unsigned int	2	0	65535	—	mV
0x3B	R	CellVoltage5	unsigned int	2	0	65535	—	mV
0x3C	R	CellVoltage4	unsigned int	2	0	65535	—	mV
0x3D	R	CellVoltage3	unsigned int	2	0	65535	—	mV
0x3E	R	CellVoltage2	unsigned int	2	0	65535	—	mV
0x3F	R	CellVoltage1	unsigned int	2	0	65535	—	mV
0x4C	R	DynamicPower	signed int	2	–32768	32767	—	10 mW
0x4D	R	ExtAveCellVoltage	unsigned int	2	0	65535	—	mV
0x4E	R	PendingEDV	unsigned int	2	0	65535	—	mV
0x4F	R	StateOfHealth	unsigned int	1	0%	100%	—	
0x50	R	SafetyAlert	hex	4+1	—	—	—	—
0x51	R	SafetyStatus	hex	4+1	—	—	—	—
0x52	R	PFAAlert	hex	2+1	—	—	—	—
0x53	R	PFStatus	hex	4+1	—	—	—	—
0x54	R	OperationStatus	hex	4+1	—	—	—	—
0x55	R	ChargingStatus	hex	2+1	—	—	—	—
0x56	R	GaugingStatus	hex	2+1	—	—	—	—
0x57	R	ManufacturingStatus	hex	2+1	—	—	—	—
0x58	R	AFESStatus	hex	2+1	—	—	—	—
0x59	R	AFEConfig	String	10+1	—	—	—	ASCII
0x5A	R	AFEVCx	String	32+1	—	—	—	ASCII
0x5B	R	AFEDData	String	13+1	—	—	—	ASCII
0x60	R	Lifetime Data Block 1	String	32+1	—	—	—	
0x61	R	Lifetime Data Block 2	String	32+1	—	—	—	
0x62	R	Lifetime Data Block 3	String	14+1	—	—	—	
0x63	R	Lifetime Data Block 4	String	20+1	—	—	—	
0x64	R	Lifetime Data Block 5	String	14+1	—	—	—	
0x65	R	Lifetime Data Block 6	String	32+1	—	—	—	
0x66	R	Lifetime Data Block 7	String	32+1	—	—	—	
0x70	R	ManufacturerInfo	String	32+1	—	—	—	
0x71	R	DASStatus 1	String	32+1	—	—	—	
0x72	R	DASStatus 2	String	32+1	—	—	—	
0x80	R	CUV Snapshot	String	32+1	—	—	—	
0x81	R	COV Snapshot	String	32+1	—	—	—	

17.2 0x00 ManufacturerAccess() and 0x44 ManufacturerBlockAccess()

The *ManufacturerAccess()* and *ManufacturerBlockAccess()* commands make available a variety of data:

- *ManufacturerAccess()* provides access to the data through the Smart Battery data set standard, including when in SEALED mode, using a sequence of a *ManufacturerAccess()* write word and a *ManufacturerData()* block read.
- The *ManufacturerBlockAccess()* is an extended command that enables access to the same data, but through a simpler block write/read sequence to the same command.

ManufacturerAccess() example to read *LifetimeDataBlock1()*:

- Send *LifetimeDataBlock1()* command through the *ManufacturerAccess()*: SMBus Write Word of 0x0060 to command 0x00.
- SMBus Read Block of command 0x23: The first two bytes of the return block will be the data length and the next bytes will be the data of the command.

ManufacturerBlockAccess() example to read *LifetimeDataBlock1()*:

- Send *LifetimeDataBlock1()* command through the *ManufacturerBlockAccess()*: SMBus Write Block of 0x60 + 0x00 to command 0x44. (Data must be sent in Little Endian.)
- SMBus Read Block of command 0x44: The first two bytes of the return block will be the Manufacturer Access command, followed by return data of the command.

Each data entity read/write through *ManufacturerBlockAccess()* is in Little Endian. For example, a 2-byte data 0x1234 should be read/write as 0x34 + 0x12; a 4-byte 0x12345678 data should be read/write as 0x78+ 0x56+ 0x34 + 0x12.

There are two exceptions:

- 0x0035 *SecurityKeys()*: This Manufacturer Access command allows the user to read or change the Unseal/Full Access keys. The above description is applied when reading the security keys. However, only the *ManufacturerBlockAccess()* can be used to change the security keys.
To write data through *ManufacturerBlockAccess()*, follow the SMBus write block protocol with the first two bytes being the *SecurityKeys()*, followed by the desired new keys' values. See [ManufacturerAccess\(\) 0x0035 Security Keys](#) for details.
- 0x0037 *AuthenticationKey()*: This Manufacturer Access command allows users to change the authentication key. Sending the new authentication key through *ManufacturerBlockAccess()* is supported. Additionally, the gauge also supports the approach of updating the authentication keys by sending the new keys to *ManufacturerInput()*. See [Section 17.2.32](#) for details.

Table 17-2. ManufacturerAccess() Command List

Command	Function	Access	Format	Data Read on 0x44 and 0x23	DATA READ on 0x2F	Not Available in SEALED Mode	Type	Unit
0x0001	DeviceType	R	Block	√	—	—	Hex	—
0x0002	FirmwareVersion	R	Block	√	—	—	Hex	—
0x0003	HardwareVersion	R	Block	√	—	—	Hex	—
0x0004	IFChecksum	R	Block	√	—	—	Hex	—
0x0005	StaticDFSsignature	R	Block	√	—	—	Hex	—
0x0006	ChemID	R	Block	√	—	—	Hex	—
0x0008	StaticChemDFSsignature	R	Block	√	—	—	Hex	—
0x0009	AllDFSsignature	R	Block	√	—	—	Hex	—
0x0010	ShutdownMode	W	—	—	—	—	Hex	—
0x0011	SleepMode	W	—	—	—	√	Hex	—
0x0012	DeviceReset	W	—	—	—	√	Hex	—
0x001B	CellBalanceToggle	W	—	—	—	√	Hex	—
0x001C	AFEDelayDisable	W	—	—	—	√	Hex	—
0x001D	SAFEToggle	W	—	—	—	√	Hex	—
0x001E	PrechargeFET	W	—	—	—	√	Hex	—
0x001F	ChargeFET	W	—	—	—	√	Hex	—
0x0020	DischargeFET	W	—	—	—	√	Hex	—
0x0021	Gauging	W	—	—	—	√	Hex	—

Table 17-2. ManufacturerAccess() Command List (continued)

Command	Function	Access	Format	Data Read on 0x44 and 0x23	DATA READ on 0x2F	Not Available in SEALED Mode	Type	Unit
0x0022	FETControl	W	—	—	—	√	Hex	—
0x0023	LifetimeDataCollection	W	—	—	—	√	Hex	—
0x0024	PermanentFailure	W	—	—	—	√	Hex	—
0x0025	BlackBoxRecorder	W	—	—	—	√	Hex	—
0x0026	SAFE	W	—	—	—	√	Hex	—
0x0027	LEDDisplayEnable	W	—	—	—	√	Hex	—
0x0028	LifetimeDataReset	W	—	—	—	√	Hex	—
0x0029	PermanentFailureDataReset	W	—	—	—	√	Hex	—
0x002A	BlackBoxRecorderReset	W	—	—	—	√	Hex	—
0x002B	LED Toggle	W	—	—	—	√	Hex	—
0x002C	LEDDisplayPress	W	—	—	—	√	Hex	—
0x002D	CalibrationMode	W	—	—	—	√	Hex	—
0x0030	SealDevice	W	—	—	—	—	Hex	—
0x0035	SecurityKeys	R/W	Block	√	—	—	Hex	—
0x0037	AuthenticationKey	R/W	Block	—	√	—	Hex	—
0x0041	DeviceReset	W	—	—	—	√	Hex	—
0x0050	SafetyAlert	R	Block	√	—	—	Hex	—
0x0051	SafetyStatus	R	Block	√	—	—	Hex	—
0x0052	PFAAlert	R	Block	√	—	—	Hex	—
0x0053	PFStatus	R	Block	√	—	—	Hex	—
0x0054	OperationStatus	R	Block	√	—	—	Hex	—
0x0055	ChargingStatus	R	Block	√	—	—	Hex	—
0x0056	GaugingStatus	R	Block	√	—	—	Hex	—
0x0057	ManufacturingStatus	R	Block	√	—	—	Hex	—
0x0058	AFESStatus	R	Block	√	—	—	Hex	—
0x0059	AFEConfig	R	String	√	—	—		—
0x005A	AFEVCx	R	String	√	—	—		—
0x005B	AFEDData	R/W	String	√	—	—		—
0x0060	LifetimeDataBlock1	R	Block	√	—	—	Mixed	Mixed
0x0061	LifetimeDataBlock2	R	Block	√	—	—	Mixed	Mixed
0x0062	LifetimeDataBlock3	R	Block	√	—	—	Mixed	Mixed
0x0063	LifetimeDataBlock4	R	Block	√	—	—	Mixed	Mixed
0x0064	LifetimeDataBlock5	R	Block	√	—	—	Mixed	Mixed
0x0065	LifetimeDataBlock6	R	Block	√	—	—	Mixed	Mixed
0x0066	LifetimeDataBlock7	R	Block	√	—	—	Mixed	Mixed
0x0070	ManufacturerInfo	R	Block	√	—	—	Hex	—
0x0071	DASStatus1	R	Block	√	—	—	Mixed	Mixed
0x0072	DASStatus2	R	Block	√	—	—	Mixed	Mixed
0x0080	CUVSnapshot	R	Block	√	—	—	Hex	—
0x0081	COVSnapshot	R	Block	√	—	—	Hex	—
0x01yy	DFAccessRowAddress	R/W	Block	—	√	√	Hex	—
0xF00	ROMMode	W	—	—	—	√	Hex	—
0xF080	ExitCalibrationOutput	R/W	Block	√	—	√	Hex	—
0xF081	OutputCellVoltageforCalibration	R/W	Block	√	—	√	Hex	—
0xF082	OutputCellVoltageCCandTemp forCalibration	R/w	Block	√	—	√	Hex	—

17.2.1 ManufacturerAccess() 0x0000 ManufacturerBlockAccess() or ManufacturerData()

A read on this command returns the lowest 16-bit of the *OperationStatus()* data.

17.2.2 ManufacturerAccess() 0x0001 Device Type

The device can be checked for the IC part number. When 0x0001 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the IC part number on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: aaAA, where:

Value	Description
aaAA	Device type

17.2.3 ManufacturerAccess() 0x0002 Firmware Version

The device can be checked for the firmware version of the IC. When 0x0002 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the firmware revision on *ManufacturerBlockAccess()* or *ManufacturerData()* in the following format: ddDDvvVVbbBBTtzzZZRREE, where:

Value	Description
ddDD	Device Number
vvVV	Version
bbBB	Build Number
ttTT	Firmware Type
zzZZ	CEDV Version
RREE	Reserved

17.2.4 ManufacturerAccess() 0x0003 Hardware Version

The device can be checked for the hardware version of the IC. When 0x0003 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the hardware revision on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*.

Value	Description
aaAA	Hardware Version

17.2.5 ManufacturerAccess() 0x0004 Instruction Flash Signature

The device can return the instruction flash signature. When 0x0004 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the IF signature on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* after a wait time of 250 ms.

17.2.6 ManufacturerAccess() 0x0005 Static DF Signature

The device can return the data flash checksum. When 0x0005 is written to *ManufacturerAccess()* the BQ78350-R1A returns the signature of all static DF on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* after a wait time of 250 ms. The MSB is set to 1 if the calculated signature does not match the signature stored in DF.

17.2.7 ManufacturerAccess() 0x0006 Chemical ID

This command returns the chemical ID of the OCV tables used in the gauging algorithm. When 0x0006 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the chemical ID on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()*.

17.2.8 ManufacturerAccess() 0x0008 Static Chem DF Signature

The device can return the data flash checksum. When 0x0008 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the signature of all static chemistry DF on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* after a wait time of 250 ms. The MSB is set to 1 if the calculated signature does not match the signature stored in DF.

17.2.9 ManufacturerAccess() 0x0009 All DF Signature

NOTE: This command is supported in the BQ78350-tbd-R2 device, not the BQ78350-R1A device. The BQ78350-R1A firmware includes some DF values that can vary from device to device. This causes the checksum to return different values on different devices even if the same .srec is loaded. Contact TI for more information.

The device can return the data flash checksum. When 0x0009 is written to *ManufacturerAccess()*, the BQ78350-R1A returns the signature of all DF parameters on a subsequent read on *ManufacturerBlockAccess()* or *ManufacturerData()* after a wait time of 250 ms. The MSB is set to 1 if the calculated signature does not match the signature stored in DF. It is normally expected that this signature will change due to updates of lifetime, gauging, and other information.

17.2.10 ManufacturerAccess() 0x0010 SHUTDOWN Mode

The device can be sent to SHUTDOWN mode before shipping to reduce power consumption to a minimum. The device will wake up when a voltage is applied to companion AFE BQ769x0. When the pack is sealed, this feature requires the command be sent twice, one after the other, for safety. Once SHUTDOWN mode is enabled, it is not possible to clear it without entering SHUTDOWN mode.

Status	Condition	Action
Normal	<i>OperationStatus()</i> [SDM] = 0	Shutdown feature is armed internally.
Arm	<i>OperationStatus()</i> [SEC1, SEC0] = [1, 1] 0x0010 sent to <i>ManufacturerAccess()</i>	Shutdown feature is armed internally.
Enable	0x0010 to <i>ManufacturerAccess()</i> when <i>OperationStatus</i> [SEC1, SEC0] = [1, 1] OR 0x0010 to <i>ManufacturerAccess()</i> when shutdown feature is armed.	<i>OperationStatus()</i> [SDM] = 1
Trip	<i>Current()</i> <= 0 AND <i>OperationStatus()</i> [SDM] = 1	No charging or discharging allowed. Device shutdown
Recovery	Voltage on companion AFE BQ769x0 TS1 pin > V _{BOOT}	Device powers up (reset)

17.2.11 ManufacturerAccess() 0x0011 SLEEP Mode

The device can be send to SLEEP mode with *ManufacturerAccess()* if the sleep conditions are met.

Status	Condition	Action
Enable	0x0011 to <i>ManufacturerAccess()</i>	<i>OperationStatus()</i> [SLEEPM] = 1
Activate	<i>Current()</i> < Power:Sleep Current	Turn off PCHG FET Turn off CHG FET if FET Options [SLEEPCHG] = 0 Device goes to sleep. Device wakes up every Power:Sleep Voltage Time period to measure voltage and temperature. Device wakes up every Power:Sleep Current Time period to measure current.
Exit	<i>Current()</i> > Configuration:Sleep Current	<i>OperationStatus()</i> [SLEEPM] = 0 Returns to NORMAL mode.
Exit	<i>SafetyAlert()</i> flag or <i>PFAAlert()</i> flag set	<i>OperationStatus()</i> [SLEEPM] = 0 Returns to NORMAL mode

17.2.12 ManufacturerAccess() 0x001B Cell Balance Toggle

This command activates/deactivates the internal cell balance FETs to ease testing during manufacturing. When 0x001B is written to *ManufacturerAccess()* when *ManufacturingStatus()[CB_TEST]* = 1, then cell balance FETs for ODD cells (Cell 1, Cell 3, Cell 5, and so on) are turned ON. If 0x001B is written to *ManufacturerAccess()* once again, then the cell balance FETs for EVEN cells (Cell 2, Cell 4, Cell 6, and so on) are turned ON. If 0x001B is written to *ManufacturerAccess()* once again, then the cell balance FETs are turned OFF and *ManufacturingStatus()[CB_TEST]* is cleared to 0.

CAUTION

This feature should not be used with cells connected to the BQ769x0.

17.2.13 ManufacturerAccess() 0x001C AFE Delay Disable

This command deactivates the companion AFE BQ769x0 output to ease testing during manufacturing. When 0x001C is written to *ManufacturerAccess()* when *ManufacturingStatus()[AFE_DD_TEST]* = 0, then the companion AFE protection delays are reduced to ~250 ms if they are longer than this in NORMAL mode and *ManufacturingStatus()[AFE_DD_TEST]* is set to 1. If 0x001C is written to *ManufacturerAccess()* once again, then the companion AFE protection delays are returned to their normal settings and *ManufacturingStatus()[AFE_DD_TEST]* is cleared to 0.

17.2.14 ManufacturerAccess() 0x001D SAFE Toggle

This command activates/deactivates the SAFE output to ease testing during manufacturing. When 0x001D is written to *ManufacturerAccess()* when *ManufacturingStatus()[SAFE_EN]* = 0, then the SAFE pin is driven high and *ManufacturingStatus()[SAFE_EN]* is set to 1. If 0x001D is written to *ManufacturerAccess()* once again, then the SAFE pin returns low and *ManufacturingStatus()[SAFE_EN]* is cleared to 0.

17.2.15 ManufacturerAccess() 0x001E PRE-CHG FET

This command turns on/off Pre-CHG FET drive function to ease testing during manufacturing. When 0x001E is written to *ManufacturerAccess()* when *ManufacturingStatus()[FET_EN,PCHG_TEST]* = 0, 0, then PCHG turns ON and *ManufacturingStatus()[PCHG_TEST]* is set to 1. If 0x001E is written to *ManufacturerAccess()* once again, then PCHG turns OFF and *ManufacturingStatus()[PCHG_TEST]* is cleared to 0.

17.2.16 ManufacturerAccess() 0x001F CHG FET

This command turns on/off CHG FET drive function to ease testing during manufacturing. When 0x001F is written to *ManufacturerAccess()* when *ManufacturingStatus()[FET_EN,CHG_TEST]* = 0, 0, then CHG turns ON and *ManufacturingStatus()[CHG_TEST]* is set to 1. If 0x001F is written to *ManufacturerAccess()* once again, then CHG turns OFF and *ManufacturingStatus()[CHG_TEST]* is cleared to 0.

17.2.17 ManufacturerAccess() 0x0020 DSG FET

This command turns on/off DSG FET drive function to ease testing during manufacturing. When 0x0020 is written to *ManufacturerAccess()* when *ManufacturingStatus()[FET_EN,DSG_TEST]* = 0, 0, then DSG turns ON and *ManufacturingStatus()[DSG_TEST]* is set to 1. If 0x0020 is written to *ManufacturerAccess()* once again, then DSG turns OFF and *ManufacturingStatus()[DSG_TEST]* is cleared to 0.

17.2.18 ManufacturerAccess() 0x0022 FET Control

This command disables/enables control of the CHG, DSG, and PCHG FET by the firmware. When 0x0022 is written to *ManufacturerAccess()* when *ManufacturingStatus()[FET_EN]* = 0, then FETs are controlled by the firmware and *ManufacturingStatus()[FET_EN]* is set to 1. If 0x0022 is written to *ManufacturerAccess()* once again, then CHG, DSG and PCHG turn OFF and *ManufacturingStatus()[FET_EN]* is cleared to 0.

17.2.19 ManufacturerAccess() 0x0023 Lifetime Data Collection

This command disables/enables Lifetime Data Collection for ease of manufacturing. When 0x0023 is written to *ManufacturerAccess()* when *ManufacturingStatus()[LF_EN]* = 0, then the Lifetime Data Collection feature is enabled and *ManufacturingStatus()[LF_EN]* is set to 1. If 0x0023 is written to *ManufacturerAccess()* once again, then the Lifetime Data Collection feature is disabled and *ManufacturingStatus()[LF_EN]* is cleared to 0.

17.2.20 ManufacturerAccess() 0x0024 Permanent Failure

This command disables/enables Permanent Failure for ease of manufacturing. When 0x0024 is written to *ManufacturerAccess()* when *ManufacturingStatus()[PF_EN]* = 0, then the Permanent Failure feature is enabled and *ManufacturingStatus()[PF_EN]* is set to 1. If 0x0024 is written to *ManufacturerAccess()* once again, then the Permanent Failure feature is disabled and *ManufacturingStatus()[PF_EN]* is cleared to 0.

17.2.21 ManufacturerAccess() 0x0025 Black Box Recorder

This command enables/disables Black Box Recorder function for ease of manufacturing. When 0x0025 is written to *ManufacturerAccess()* when *ManufacturingStatus()[BBR_EN]* = 0, then the Black Box Recorder feature is enabled and *ManufacturingStatus()[BBR_EN]* is set to 1. If 0x0025 is written to *ManufacturerAccess()* once again, then the Black Box Recorder feature is disabled and *ManufacturingStatus()[BBR_EN]* is cleared to 0.

17.2.22 ManufacturerAccess() 0x0026 SAFE

This command disables/enables firmware-based SAFE pin activation to ease testing during manufacturing. When 0x0026 is written to *ManufacturerAccess()* when *ManufacturingStatus()[SAFE_EN]* = 0, then the SAFE pin is enabled and *ManufacturingStatus()[SAFE_EN]* is set to 1. If 0x0026 is written to *ManufacturerAccess()* once again, then the SAFE pin is disabled and *ManufacturingStatus()[SAFE_EN]* is cleared to 0.

17.2.23 ManufacturerAccess() 0x0027 LED Display Enable

This command enables or disables the LED display function from being triggered by the $\overline{\text{DISP}}$ pin to ease testing during manufacturing. The initial setting is loaded from *Mfg Status Init[LED_EN]*. If the *ManufacturingStatus()[LED_EN]* = 0, sending this command will enable the LED display and the *ManufacturingStatus()[LED_EN]* = 1 and vice versa. In UNSEALED mode, the *ManufacturingStatus()[LED_EN]* status is copied to *Mfg Status Init[LED_EN]* when the command is received by the gauge. The device remains on its latest setting prior to a reset.

17.2.24 ManufacturerAccess() 0x0028 Lifetime Data Reset

This command resets Lifetime data in data flash for ease of manufacturing. When 0x0028 is written to *ManufacturerAccess()*, the Lifetime data stored in data flash is cleared.

Status	Condition	Action
Reset	0x0028 to <i>ManufacturerAccess()</i>	Clear Lifetime Data in DF

17.2.25 ManufacturerAccess() 0x0029 Permanent Fail Data Reset

This command resets PF data in data flash for ease of manufacturing. When 0x0029 is written to *ManufacturerAccess()*, the Permanent Failure data stored in data flash is cleared.

Status	Condition	Action
Reset	0x0029 to <i>ManufacturerAccess()</i>	Clear PF Data in DF

17.2.26 *ManufacturerAccess()* 0x002A Black Box Recorder Reset

This command resets the Black Box Recorder data in data flash for ease of manufacturing. When 0x002A is written to *ManufacturerAccess()*, the Black Box Recorder data stored in data flash is cleared.

Status	Condition	Action
Reset	0x002A to <i>ManufacturerAccess()</i>	Clear Black Box Recorder data in DF

17.2.27 *ManufacturerAccess()* 0x002B LED TOGGLE

This command toggles the LED display from off to on and from on to off to help streamline testing during manufacturing. The status of the display is indicated through *OperationStatus()[LED]*.

17.2.28 *ManufacturerAccess()* 0x002C LED Display Press

This command simulates a low-high-low detection of the $\overline{\text{DISP}}$ pin, activating the LED display according to the LED Support data flash setting. This command forces RSOC to 100% in order to demonstrate all LEDs in use, the full speed, and the brightness.

17.2.29 *ManufacturerAccess()* 0x002D CALIBRATION Mode

This command disables/enables entry into CALIBRATION mode. When 0x002D is written to *ManufacturerAccess()* when *ManufacturingStatus()[CAL_EN]* = 0, then *ManufacturingData()* is enabled to output ADC and CC raw data, is controllable with 0xF081 and 0xF082 on *ManufacturerAccess()*, and *ManufacturingStatus()[CAL_EN]* is set to 1. If 0x002D is written to *ManufacturerAccess()* once again, then *ManufacturingData()* returns default data and *ManufacturingStatus()[CAL_EN]* is cleared to 0.

17.2.30 *ManufacturerAccess()* 0x0030 Seal Device

This command seals the device for the field, disabling certain SBS commands and access to DF. When 0x0030 is written to *ManufacturerAccess()* when *OperationStatus()[SEC1, SEC0]* = 0,1 or 1,0, then the BQ78350-R1A device enters SEALED mode and *OperationStatus()[SEC1, SEC0]* is set to 1,1.

17.2.31 *ManufacturerAccess()* 0x0035 Security Keys

This is a read/write command that changes the Unseal and Full Access keys. To read the keys, sending the *SecurityKeys()* command to either the *ManufacturerAccess()* 0x00 or the *ManufacturerBlockAccess()* 0x44 returns the keys on *ManufacturerBlockAccess()* or *ManufacturerData()*.

To change the keys, the write operations must send through *ManufacturerBlockAccess()* 0x44 with *SecurityKeys()* followed by the keys. Each parameter word entry must be sent in Little Endian.

Example:

Changing the Unseal key to 0x0123, 0x4567 and the Full Access key to 0x89AB, 0xCDEF:

SMBus write block: command = 0x44, block = 0x35 + 0x00 + 0x23 + 0x01 + 0x67 + 0x45 + 0xAB + 0x89 + 0xEF + 0xCD

Byte0–1: *SecurityKeys()* command

Byte2–3: Unseal Key LSW

Byte4–5: Unseal Key MSW

Byte6–7: Full Access Key LSW

Byte8–9: Full Access Key MSW

17.2.32 *ManufacturerAccess()* 0x0037 Authentication Key

This command enters a new authentication key into the device.

Status	Condition	Action
Initiate	<i>OperationStatus()</i> [SEC1, SEC0] = 0,1 AND 0x0037 to <i>ManufacturerAccess()</i>	<i>OperationStatus()</i> [AUTH] = 1 160-bit random number available at <i>ManufacturerInput()</i>
Enter Key	Correct 128-bit Key written to <i>ManufacturerInput()</i> in the format 0xAABBCCDDEEFFGGHHIIJJKLLMMNNOOPP, where AA is LSB.	Wait time 250 ms <i>OperationStatus()</i> [AUTH] = 0 Device returns 160-bit HMAC digest at <i>ManufacturerInput()</i> in the format 0xAABBCCDDEEFFGGHHIIJJKLLMMNNOOPPQQRRSSTTT, where AA is LSB. The HMAC digest was calculated using the random number + key. Compare with own calculations, check the validity of the key.

17.2.33 ManufacturerAccess() 0x0041 Device Reset

This command resets the device. When 0x0012 or 0x0041 is written to *ManufacturerAccess()*, the BQ78350-R1A is reset.

17.2.34 ManufacturerAccess() 0x0050 SafetyAlert

This command returns the *SafetyAlert()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

Status	Condition	Action
Activate	0x0050 to <i>ManufacturerAccess()</i>	Outputs <i>SafetyAlert()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i>

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	OC	CTOS	RSVD	PTOS	RSVD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	OCDL	OTF	AFE_OVRD	UTD	UTC	OTD	OTC	ASCDL	ASCD	AOLDL	AOLD	OCD	OCC	COV	CUV

RSVD (Bits 31–21): Reserved

OC (Bit 20): Overcharge

1 = Detected

0 = Not Detected

CTOS (Bit 19): Charge Timeout Suspend

1 = Detected

0 = Not Detected

RSVD (Bit 18): Reserved

PTOS (Bit 17): Precharge Timeout Suspend

1 = Detected

0 = Not Detected

RSVD (Bits 16–15): Reserved

OCDL (Bit 14): Overcurrent During Discharge Latch

1 = Detected

0 = Not Detected

OTF (Bit 13): Overtemperature Fault

1 = Detected

0 = Not Detected

AFE_OVRD (Bit 12): AFE Alert

1 = Detected

0 = Not Detected

UTD (Bit 11): Undertemperature During Discharge

1 = Detected

0 = Not Detected

UTC (Bit 10): Undertemperature During Charge

1 = Detected

0 = Not Detected

OTD (Bit 9): Overtemperature During Discharge

1 = Detected

0 = Not Detected

OTC (Bit 8): Overtemperature During Charge

1 = Detected

0 = Not Detected

ASCDL (Bit 7): Short Circuit During Discharge Latch

1 = Detected

0 = Not Detected

ASCD (Bit 6): Short Circuit During Discharge

1 = Detected

0 = Not Detected

AOLDL (Bit 5): Overload During Discharge Latch

1 = Detected

0 = Not Detected

AOLD (Bit 4): Overload During Discharge

1 = Detected

0 = Not Detected

OCD (Bit 3): Overcurrent During Discharge

1 = Detected

0 = Not Detected

OCC (Bit 2): Overcurrent During Charge

1 = Detected

0 = Not Detected

COV (Bit 1): Cell Overvoltage

1 = Detected

0 = Not Detected

CUV (Bit 0): Cell Undervoltage

1 = Detected

0 = Not Detected

17.2.35 ManufacturerAccess() 0x0051 SafetyStatus

This command returns the *SafetyStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.